Economic Evaluations of Food is Medicine Programs: A Literature Review

Katherine Walcott

College for Public Health and Social Justice, Saint Louis University

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Dr. Nancy Weaver

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Abstract

Introduction: Food is Medicine (FIM) addresses chronic, diet-related disease and food insecurity by incorporating nutrition services into health care delivery. While evidence suggests FIM improves health outcomes and reduces health care utilization, its cost effectiveness is less well understood.

Methods: A literature review examined economic evaluations of FIM interventions, including medically tailored meals (MTMs), produce prescriptions (PRx), and incentive programs. Six databases were searched in March 2026, with additional results captured through snowballing. Studies were included if they conducted any form of economic evaluation of FIM, with both peer-reviewed and grey literature considered.

Results: Fifteen studies met inclusion criteria, including six simulation and nine observational studies. Populations, FIM approaches, and economic evaluation methods varied widely, though FIM was associated with reductions in health care utilization and costs. Simulations generally demonstrated cost effectiveness, particularly over long time horizons, with many incremental cost-effectiveness ratios (ICERs) falling below commonly accepted willingness-to-pay (WTP) thresholds. Observational studies more frequently reported immediate cost savings.

Discussion and Conclusion: Overall, FIM demonstrates potential to improve health outcomes and reduce health care costs. However, heterogeneity in study design limits direct comparability. Despite promising findings, significant gaps remain, including real-world economic evaluations of government-sponsored FIM programs. Expanding rigorous, standardized economic research may help inform policy decisions and lead to broader adoption of FIM interventions.

Keywords: Food Is Medicine, cost effectiveness, economic evaluation, medically tailored meals; produce prescriptions, health care costs

Introduction

Every year in the United States (US), poor diet is responsible for over 600,000 deaths, and diet-related conditions are responsible for $1.1 trillion in health care spending and lost productivity.[1] These conditions, which include diabetes, obesity, hypertension, cardiovascular diseases, liver diseases, kidney diseases, and certain types of cancers,[2] disproportionately affect racial and ethnic minorities, individuals of low socioeconomic status,[1,3] and individuals with low levels of education.[4] In addition to facing health disparities, these populations also face higher rates of food insecurity[1,3]—defined as the limited or uncertain ability to obtain enough safe, nutritious food due to economic or social conditions[5]—and other barriers to shopping for and preparing healthy food such as physical disability, living in food deserts, and time constraints.[6]

This creates a vicious cycle whereby the inability to afford and access healthy food not only contributes to the development of chronic diet-related conditions[7] but also to difficulties managing them. This cycle is further compounded by challenges affording health care[8,9] (Figure 1), a problem heightened for those living in the US, which spent $14,885 per capita on health care in 2024 compared to its peers' average of $7,371.[9] Though this challenge is shared by all Americans, it is disproportionately shouldered by the same populations that face heightened health disparities and food insecurity.[10] Solving this disproportionate burden is necessary not only from a moral and societal perspective, but also from an economic perspective, given these individuals make up a disproportionate share of health care spending.[8,11–13]

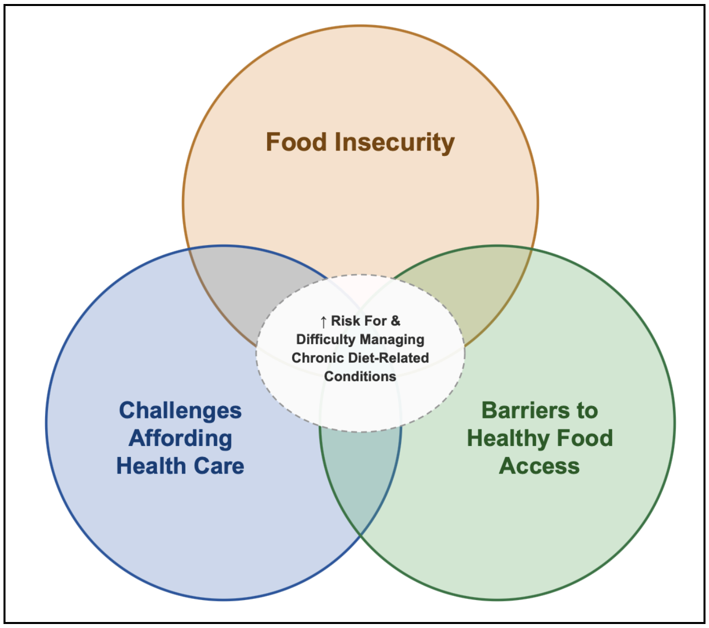

Figure 1: The Triple Burden of Food Insecurity, Challenges Affording Health Care, and Barriers to Healthy Food Access
Image generated using Claude[14]

Over the years, numerous attempts have been made to address this triple burden. However, because the root causes of the increase in diet-related conditions are rooted in the environments in which people live,[15] attempts have been met with varying levels of success. Nevertheless, one of the more recent and successful attempts has been Food Is Medicine (FIM). Though it is not a silver bullet in the fight against diet-related disease—a fight which will need to include multiple solutions across the policy, health care, and community settings—FIM offers much promise as a way to improve food quality, food access, and health while reducing health disparities and costs.

What is Food is Medicine

FIM approaches address the burdens described above by providing healthy, low or no-cost food to individuals with diet-related conditions who may face food insecurity. Because racial and ethnic minorities, individuals of low socioeconomic status, and individuals with low levels of education are disproportionately more likely to be diagnosed with diet-related diseases, FIM is also a valuable tool for advancing health equity.[16] FIM is relatively new concept, with approaches such as medically tailored meals (MTMs) growing out of the Ryan White Comprehensive AIDS Resources Act of 1990.[17] Today, due to data showing improvements in consumption of healthy food, food insecurity, health outcomes, and health care costs, hospitals, insurers, communities, governments,[1] and even retailers[18] are increasingly turning to FIM.[1]

In the current landscape, some stakeholders use the term FIM, others use the term Food As Medicine (FAM), and others use them interchangeably.[19] However, there are subtle distinctions. The term FIM is more frequently used by health insurers, health systems, and funders in the health care space to describe reimbursable food-based interventions integrated into the health care system, whereas the term FAM is used more frequently by nutrition professionals, public health organizations, and researchers to refer to a broader approach to integrating food and health.[20] Because this paper focuses on specific food-based interventions integrated into the health care system, the term FIM will be used.

FIM approaches can take many forms, including MTMs, medically tailored groceries (MTGs), medically supportive meals (MSMs), medically supportive groceries (MSGs), produce prescriptions (PRx), incentive programs, disincentive programs, individual or group nutrition counseling or education, or a combination of multiple approaches (Table 1, Figure 2).[1,3,18] In all cases, health care providers or payers are involved to identify patients eligible for FIM approaches, such as patients with diet-related conditions and food insecurity.[3] While FIM encompasses a broad set of interventions, current literature largely focuses on MTMs, MTGs, and PRx. Accordingly, this paper will primarily focus on these interventions and incorporate evidence from other approaches where available.

Table 1: Food Is Medicine Approaches and Definitions

FIM Approach	**Abbreviation**	**Definition**
Medically tailored meals	MTMs	Prepared meals, often delivered directly to patients' homes, developed by Registered Dietitians (RDs) to meet the specific needs of patients with certain diet-related conditions[1]
Medically tailored groceries	MTGs	Unprepared healthy foods, which may be delivered directly to patients' homes or provided indirectly via voucher[1,21]
Medically supportive meals	MSMs	Similar to medically tailored meals; however, these are typically designed for general conditions rather than specific individuals[22]
Medically supportive groceries	MSGs	Similar to medically tailored groceries; however, these are typically designed for general conditions rather than specific individuals[22]
Produce prescriptions	PRx	Vouchers, physical cards, or electronic cards prescribed by health care providers that can be exchanged for fruits or vegetables[1]
Incentive programs	–	Programs that provide financial means of encouragement to purchase healthier foods, for example providing additional thirty cents in monetary food assistance for every dollar spent on produce[22]
Disincentive programs	–	Programs that provide financial means of discouragement to purchase unhealthy foods, for example adding thirty cents to the cost of every dollar spent on unhealthy foods such as sugar-sweetened beverages[23]
Individual or group nutrition education	–	General instruction provided by a health care professional, such as a RD, on how to manage nutrition-related problems[24]
Individual or group nutrition counseling	–	Personalized and interactive guidance provided by a health care professional, such as a RD, on how to manage nutrition-related problems[24]

Impacts of Food is Medicine

FIM interventions have demonstrated the ability to increase healthy food consumption, reduce food insecurity, improve health outcomes, and reduce health care costs. In their 2025 annual report of the FIM landscape, the American Heart Association (AHA) conducted reviews of the evidence of MTMs, MTGs, and PRx.[21] PRx were associated with increases in fruit and vegetable intake of 0.5–1 servings/day, especially when paired with supportive measures such as education, and MTMs were associated with additional nutritional improvements in older adults and individuals with heart failure.[21] MTMs, MTGs, and PRx were also all associated with improvements in food insecurity.[21]

In terms of health outcomes, MTMs were associated with improvements in blood pressure and body mass index (BMI), MTGs with improvements in blood pressure, BMI, and hemoglobin A1c (HbA1c), and PRx with a reduction in body weight of 2.4%–4.7%, depending on the length of the intervention.[21] Additionally, one study of MTMs reported significant decreases in emergency department and inpatient visits, with a per patient cost savings of over $12,000, and one study of MTGs reported a significant reduction in emergency department visits.[21] Furthermore, MTMs have been "associated with a 37% to 52% lower risk of hospitalization, 16% to 31% reduction in monthly health care expenditures, and decreased net costs of approximately $2,500 per patient-year."[17]

However, despite the positive findings of FIM on healthy food consumption, food insecurity, health outcomes, and health care costs, less is known about their cost effectiveness. In fact, the AHA notes this area is one of the significant remaining gaps in the evidence surrounding FIM.[25] Therefore, this review will focus on summarizing the existing evidence surrounding cost effectiveness.

Economic Theory Behind Cost Effectiveness

When discussing the cost effectiveness of any intervention, it is important to ground the work in theory. Welfare economics assesses the well-being of a society by evaluating its economic activities and resource allocation.[26] The main goal is to allocate finite resources, such as money, most effectively to improve the common good.[26] To evaluate how well an activity or intervention accomplishes this goal, tools such as economic evaluation are used. Specifically, economic evaluation compares the costs and outcomes of different approaches to determine which produces the best results given resource constraints.[27]

Economic evaluation can take many forms, including cost-minimization analysis (CMA), cost-effectiveness analysis (CEA), cost-utility analysis (CUA), cost-benefit analysis (CBA), and other, less formal approaches (Table 2). From a health care perspective, institutions are expected to adopt interventions that most effectively maximize health outcomes considering their costs. Because of this, although FIM approaches show promise in addressing the diet-related conditions, they will not be widely adopted until they can demonstrate robust cost effectiveness.

Table 2: Economic Evaluation Methods and Definitions

Economic Evaluation Method	**Abbreviation**	**Definition**
Cost-minimization analysis	CMA	Interventions that achieve the same outcomes are compared to determine which incurs lower costs.[27] Whichever intervention incurs less costs is considered more cost effective by maximizing enhancements to the common good using less finite resources.
Cost-effectiveness analysis	CEA	The costs incurred by an intervention are considered with respect to the improvements seen. Improvements must take the form of natural units, for example years of life gained, instances of illness avoided, or reduction in certain laboratory values.[27] Whichever intervention achieves the same outcome for less costs is considered more cost effective.
Cost-utility analysis	CUA	The costs incurred by an intervention are considered with respect to improvements seen, but these improvements must take the form of quality-adjusted life years (QALYs), disability-adjusted life years (DALYs), or health-years equivalent.[27] Whichever intervention achieves the same outcome for less costs is considered more cost effective.
Cost-benefit analysis	CBA	All outcomes are converted to costs, allowing for a direct monetary comparison of costs to benefits.[27] When comparing interventions, whichever has the largest net benefits is considered the most cost effective; however, certain outcomes, such as the value of a human life, are more difficult to convert to monetary values.
Cost-savings analysis	CSA	Costs and benefits are compared more generally, often ignoring effects that cannot be easily converted into monetary units. This form of analysis is considered less sound than CMA.[28]
Cost-avoidance analysis	CAA	The negative outcomes avoided by an intervention are considered with particular focus on the costs these outcomes would have incurred. This form of analysis is typically used when interventions are preventative and outcomes are difficult to measure.[29]
Cost-consequence analysis	CCA	The costs incurred by an intervention are considered with respect to the improvements seen. However, costs and benefits can take the form of natural units, and a variety of costs and benefits may be compared at the same time. This form of analysis typically makes direct comparisons difficult due to a lack of a common unit.[29]
Budget impact analysis	BIA	Expenditures that are likely to result from an intervention are estimated. This form of analysis is typically conducted to assess feasibility and predict budget needs.[30]

Methods

Search Strategy

The search strategy used for this literature review was designed in consultation with a medical librarian for the Saint Louis University (SLU) College of Public Health and Social Justice. Detailed information on the search strategy can be seen in the Appendix (Table A1). Databases were searched at the beginning of March, 2026. Articles were included if they included a cost analysis of FIM interventions. For review papers, articles that included a cost analysis of FIM interventions were extracted, with all other articles in the review excluded. Articles were excluded if they were not about FIM, did not include cost analysis, were published before the year 2000, were study protocols, were magazine or newspaper articles, were abstracts for conferences, were not in English, were erratums, could not be found by SLU medical librarians, or did not separate FIM findings from those of other health-related social needs. Due to the relatively limited amount of peer-reviewed cost analyses on FIM interventions, grey literature was included, with additional articles captured via snowballing.

All database results were imported into Google Sheets. Duplicates were identified, and titles and abstracts were screened to determine eligibility. When titles and abstracts did not provide enough information on the type of intervention or economic analysis, the full-text was reviewed. After identifying all included articles, findings were extracted for analysis (Figure 3).

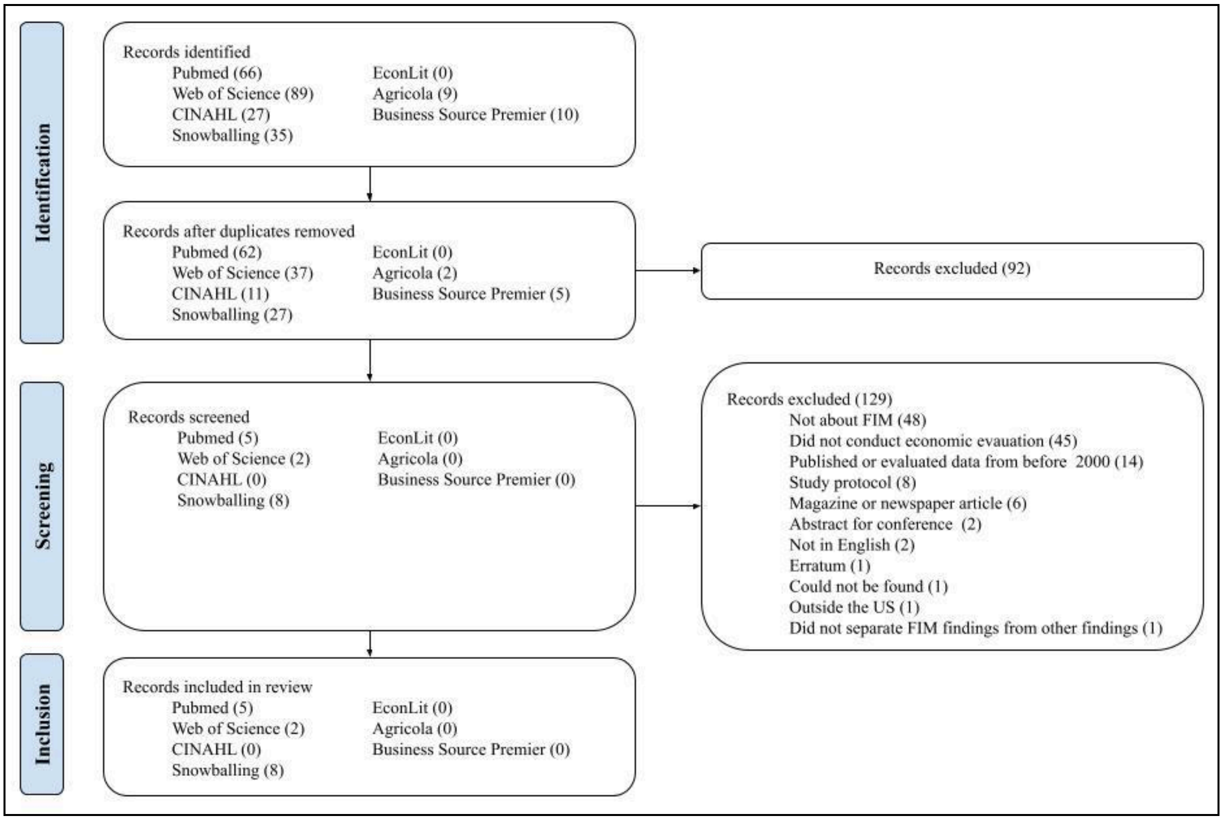

Figure 3. PRISMA diagram

Results

The search identified 236 results, of which 92 were duplicates and excluded. Of the remaining 144, 15 met eligibility criteria. Of the 15, six studies were simulation studies, and nine studies were observational studies. Three conducted CUA, three CEA, one CBA, four CSA, three CAA, and one CCA. Three studies evaluated PRx, ten evaluated MTMs, one evaluated an incentive program, and one evaluated a combination of FIM interventions. Seven studies analyzed findings in the entire US, while the rest analyzed findings in specific states (including California, Georgia, Maine, Massachusetts, Michigan, North Carolina, Ohio, Oklahoma, Pennsylvania, Rhode Island, and Texas).

Study populations varied widely in terms of race, ethnicity, insurance status, and health; however, the majority of studies examined populations with diet-related conditions and food insecurity. Additionally, despite their differences, nearly all studies examined similar age groups, ranging from individuals in their fifties to seventies. In addition to varying significantly depending on the population, results varied significantly depending on the type of FIM intervention, intervention intensity, time horizon, and economic evaluation method, limiting direct comparability across studies.

Regardless of study design, a similar theme appeared across all studies. FIM interventions were generally associated with decreases in health care costs and metrics such as admissions, readmissions, emergency department visits, and hospitalizations. Though findings differ in their robustness (Appendix Table A2), all studies suggested FIM interventions are a valuable tool to prevent poor health outcomes and generate cost savings. Simulation studies consistently demonstrated cost effectiveness, especially over medium- to long-term time horizons, while observational studies more frequently reported immediate decreases in health care costs and health care utilization. Among studies conducting CUA, ICERs varied significantly depending on the type of FIM intervention, population, and time horizon. Lower ICERs were generally observed in longer time horizons, while higher ICERs were generally observed in shorter-term interventions. Despite this variation, the majority of ICERs fell under common willingness-to-pay thresholds (WTP) of $150,000/QALY.

A summary of results has been displayed in Table 3, with more details available in the Appendix (Table A2).

Table 3. Summary of results of systematic review

Authors, Year	**Study Design, Type of Economic Evaluation**	**Study Location, Year**	**Study Population**	**Intervention**	**Results**
Simulation Studies					
Wang et al., 2023[3]	Microsimulation modeling study, CUA	US, +5, +10, and +25 yr from analysis	6.5 million food-insecure US adults (DOC-M model)[4] - Mean age: 58.2 ± 10.2 yr - Health: 100.0% DM, 74.8% HTN, 30.0% CVD	PRx ($42/mo/box or voucher)	Over 25 yr (lifetime) - Cost $37.3 billion in food and $6.99 billion in administrative costs - Save $39.6 billion in health care and $4.8 billion in productivity costs - Net societal cost savings of $50 million - ICER = $18,100/QALY Over 10 yr - Health care ICER = $58,000/QALY and societal ICER = $28,700/QALY Over five yr - Health care ICER = $92,700/QALY and societal ICER = $66,100/QALY
Wang et al., 2025[31]	Microsimulation modeling study, CUA	US, +5 and +10 yr from analysis	5,555,000 food-insecure US adults (DOC-M model)[4] - Mean age: 57–62 yr (varied by state) - Health: 100.0% DM, 19.3%–37.8% CVD (varied by state)	PRx ($47/mo/box or voucher)	Over 10 yr - Nationwide costs of $26.1 billion and health care savings of $29.2 billion - ICER <$50,000/QALY in 49/50 states - Net health care savings projected in 43/50 states, with net nationwide health care savings of $2.68 billion - Net societal savings projected in 48/50 states, with net nationwide societal savings of $4.17 billion Over five yr - Similar patterns of health benefits and economic outcomes, but cost savings decreased to 40/50 states
Hager et al., 2022[17]	Cohort simulation modeling study, CEA	US, +1 and +10 yr from analysis	6,309,998 US adults - Mean age: 68.1 ± 16.6 yr - Health: 100.0% IADL limitations, 44.9% DM, 36.5% other heart disease, 37.2% cancer, 36.1% stroke, 26.8% CHF, 21.3% MI	10 MTMs/wk for 8 months/yr ($9.30/meal)	Over 10 yr - Program costs of $298.7 billion - Reduce health care expenditures by $484.5 billion - Net health care savings of $185.1 billion Over one yr - Program costs of $24.8 billion - Reduce health care expenditures by $38.7 billion - Net health care savings of $13.6 billion
Deng et al., 2025[32]	Cohort simulation modeling study, CEA	US, +1 and +5 yr from analysis	10,358,159 US adults - Mean age: 67.8 ± 16.5 yr - Health: 100.0% 1+ IADL limitations, 74.9% HTN, 38.4% DM, 33.8% lung disease, 26.9% CHD, 21.9% cancer, 20.0% stroke	10 MTMs/wk for 8 months/yr ($11.15/meal)	Over five yr - Net nationwide policy cost savings of $111.1 billion Over one yr - Nationwide program costs of $37.3 billion - Nationwide health care cost savings of $61.0 billion - Net nationwide policy cost savings of $23.7 billion - Net per patient health care savings of $2,287

Lee et al., 2019[33]	Microsimulation modeling study, CUA	US, +5, +10, +20, and +lifetime from analysis	82,000,000 US adults (CVD-PREDICT model)[34]
			- Mean age: 68.1 ± 11.4 yr
			- Health: 57.3% HTN, 20.2% DM, 9.8% MI, 8.3% stroke, 5.9% angina
				30% subsidy on fruits and vegetables (FV), (~ $110/ person/yr)
				
				30% subsidy on fruits, vegetables, whole grains, nuts, seeds, seafood, and plant oils (FV+), (~$185/person/ yr)
					Over a lifetime
					- FV incentive
					  - Cost $7.1 billion in administrative and $115.5 billion in food subsidy costs
					  - Health care savings of $39.7 billion
					  - Net costs of $83.5 billion
					    - ICER of $18,184/QALY
					  - Net societal costs of $68.8 billion
					    - Societal ICER of $14,576/QALY
					- FV+ incentive
					  - Cost $12.2 billion in administrative and $198.2 billion in food subsidy costs
					  - Health care savings of $100.2 billion
					  - Net costs of $111.1 billion
					    - ICER of $13,194/QALY
					  - Societal costs of $80.5 billion
					    - Societal ICER of $9,497/QALY
					Over 10 yr
					- FV+ incentive
					  - ICER of $35,246/QALY in dual-eligible adults, $34,721/QALY in Medicare adults, and $58,888/QALY in Medicaid adults
					Over five yr
					- FV+ incentive
					  - ICER of $77,537/QALY in dual-eligible adults, $78,715/QALY in Medicare adults, and $135,093/QALY in Medicaid adults

Hayes et al., 2019[35]	Cohort simulation modeling study, CBA	US, +1 year from analysis	575,408 food-insecure Medicare FFS beneficiaries - Age: 25% <65 yr, 14% 65–75 yr, 61% ≥75 yr - Health: 100.0% 1+ ADL deficits, 100.0% 2+ of the following: CHF, depression, diabetes, emphysema, asthma, COPD, Alzheimer's, dementia, osteoporosis, other heart condition, paralysis, Parkinson's, rheumatoid arthritis, stroke	7 days of home-delivered MTMs ($175.98/person/wk)	Readmissions - 9,719 fewer readmissions Costs - Total costs of $101,258,974 - Gross savings due to decreased readmissions of $158,606,687 - Net savings of $57,347,713 - Savings to cost ratio of 1:57 - Additional savings likely from decreased ED and SNF admissions
Observational Studies					
Raso, 2020[36]	Retrospective observational cohort study (thesis), CEA	Detroit, MI, 2019	171 patients of the CHASS center - Mean age: 49 yr - Health: 100.0% DM	16–18 weeks of PRx, cooking demonstrations, healthy recipes, and	Costs - Total cost of $59,152.66/year Cost effectiveness - ACER of $1,901/unit change in HbA1c - Decrease in HbA1c of 0.532

				nutrition education	
Thomas et al., 2013[37]	Observational ecological study with modeled projections, CSA and BIA	US, 2009	392,594 additional adults eligible for MOW	Increasing MOW recipients by one percent	- Decrease of 0.2% in states' low-care nursing home population - Decrease of 1,722 low-care, dually eligible elderly adults in nursing homes - Initial costs of $117,568,07 in Title III-C2 expenditures - Initial Medicaid savings of $109 million - Annual state and federal savings in 26/50 states and annual costs in 22/50 states - Savings increased to 46/50 states with FL and CA excluded
Berkowitz et al., 2019[6]	Retrospective observational cohort study, CAA	MA, 2016–2019	1,020 adults (499 recipients, 521 nonrecipients) - Mean age: 52.7 ± 14.5 yr - Health: 37.5% cancer, 28.7% CHF, 28.0% ESRD, 27.3% DM, 17.6% disability, 16.2% HIV-positive	10 home-delivered MTMs/wk ($350/person/wk)	Inpatient and SNF admissions - MTMs associated with significantly fewer inpatient admissions and fewer estimated admissions - MTMs associated with fewer SNF admissions and fewer estimated admissions Costs - MTMs associated with 16% lower health care costs (decrease of $712/mo)

Berkowitz et al., 2018[38]	Retrospective cohort study, CAA	MA, 2014–2016	3,257 adults members of the CCA
			- 133 MTM recipients
			  - Mean age: 57.4 ± 8.4 yr
			  - Health: comorbidity index 0.26 ± 0.39
			- 1,002 MTM matched controls
			  - Mean age: 57.9 ± 5.4 yr
			  - Health: comorbidity index 0.26 ± 0.25
			- 624 N-MTM recipients
			  - Mean age: 73.5 ± 7.5 yr
			  - Health: comorbidity index 0.17 ± 0.32
			- 1,318 N-MTM matched controls
			  - Mean age: 73.1 ± 5.9 yr
			  - Health: comorbidity index 0.17 ± 0.26
				10 home-delivered MTMs/wk (MTMs) for 6 months ($350/person/wk)
				
				10 home-delivered non-MTMs/wk (N-MTMs) for 6 months ($146/person/wk)
					Health care utilization
					- MTMs associated with fewer ED visits, inpatient admissions, and emergency transportation use
					- N-MTMs associated with fewer ED visits and emergency transportation use but not fewer inpatient admissions
					Costs
					- MTMs associated with lower health care costs (decrease of $570/mo)
					- MTM net savings of $220/mo
					- N-MTMs associated with lower health care costs (decrease of $156/mo)
					- N-MTM net savings of $10/mo
					- Findings remained robust in sensitivity analyses

Gurvey et al., 2013[39]	Retrospective cohort study, CSA	Philadelphia, PA, 2008–2010	698 adult clients of the MANNA
			- 65 recipients
			  - Mean age: 52 ± 6.2 yr
			  - Health: 43% AIDS/HIV, 38% chronic pulmonary disease, 26% cancer, 25% mild liver disease, 25% DM, 20% CHF, 20% renal disease
			- 633 nonrecipients
			  - Mean age: 51 ± 1.2 yr
			  - Health: 47% AIDS/HIV, 35% chronic
				21 MTMs/wk + MNT, nutrition counseling, and meal planning from RDs	Health care utilization
					- Significantly lower average monthly inpatient visits and LOS among recipients (difference of 0.2 visits/mo and 6.4 days)
					- Significantly higher average monthly ED visits among recipients (difference of 0.3/mo)
					- Significantly higher percent of individuals discharging to home among recipients (difference of 21%)
					Costs
					- Significantly lower average monthly health care costs among recipients (difference of $12,638/mo)

			pulmonary disease, 31% cancer, 23% mild liver disease, 22% DM, 19% CHF, 17% renal disease		- Significantly lower average monthly inpatient costs among recipients (difference of $87,198/mo) - Higher average monthly ED visit costs among recipients (difference of $1,193/mo)
Martin et al., 2018[40]	Quasi-experimental study, CSA	ME, 2013–2015	622 adult MMC patients - <u>Mean age</u>: 71.7 ± 13.1 yr - <u>Health</u>: 10.8% principal diagnosis HF, 6.6% atrial fibrillation/dysrhythmias, 5.0% septicemia, 4.8% acute MI, 4.2% coronary atherosclerosis, 14.8% other	7 frozen MTMs/wk ($70/person/wk) + 4 wk of tools and skills training 4 wk of tools and skills training (N-MTMs)	Readmissions - Lower 30-day readmission rate among MTM recipients (difference of 2.0%) Costs - Cost savings of $212,160 - Return on investment of 387%
Shan et al., 2017[41]	Retrospective cohort study, CAA	CA, GA, NC, OK, RI, TX; 2009–2014	14,019 Medicare FFS beneficiaries	MOW	Health care utilization - Hospitalization significantly decreased by 38.9%, 38.0%, and 31.0% at 30, 90, and 180 days post MOW enrollment - ED visits significantly decreased by 28.2%, 20.7%, and 12.7% at 30, 90, and 180 days post MOW enrollment - Nursing home admissions significantly decreased by 27.8%, 37.4%, and 25.3% at 30, 90, and 180 days post MOW enrollment Health care costs - Average per person hospitalization costs significantly decreased by $361.50, $1,154.80, and $1,355.60 at 30, 90, and 180 days post MOW enrollment - Average per person SNF costs significantly decreased by $244.30, $651.80, and $363.20 at 30, 90, and 180 days post MOW enrollment - Average per person ED costs significantly decreased by $22.36 and $42.57 at 30 and 90 days post MOW enrollment and decreased by $27.47 at 180 days post MOW enrollment
Haddad et al., 2025[42]	Single-arm prospective cohort feasibility study, CSA	OH, 2021–2022	60 Cleveland Clinic patients - <u>Median age</u>: 63.5 yr (range 51–81) - <u>Health</u>: 100.0% CHF, DM, or HTN	14 home-delivered frozen MTMs/wk + additional milk, fruit, and bread for 3 months	Health care utilization - Significant decrease, increase (not significant), decrease (not significant), or no change in ED visits depending on time frame (e.g., significant decrease 60 days before vs. 60 days into MTMs, but increase 30 days before vs. 30 days into MTMs) - Significant decrease, increase (not significant), or decrease (not significant) in inpatient LOS depending on time frame Costs - Average health care savings of $12,046/patient (from 180 days before to

					180 days after MTMs) - Cost savings achieved in 37/60 patients, with average cost savings of $29,172/patient - Total annual program costs of $189,000

Hager et al., 2025[43]	Retrospective cohort study, CCA	MA, 2020–2023	22,511 food-insecure MassHealth members eligible for nutrition-related benefits via the Flexible Services Program
			- 15,557 adult and 4,846 child recipients
			  - Mean age: 44.7 ± 12.6 yr (adults), 9.4 ± 4.9 yr (children)
			  - Health: 62.9% anxiety disorder, 45.5% HTN, 45.4% obesity, 45.4% other mental health conditions, 34.1% DM, 32.4% tobacco use, 31.1% disability, 30.8% depression (adults); 34.4% obesity (children)
			- 1,497 adult and 611 child nonrecipients (eligible but did not enroll)
			  - Median age: 43.2 ± 13.2 yr (adults), 9.5 ± 4.9 yr (children)
			  - Health: 61.5% anxiety disorder, 46.2% other mental health conditions, 40.5% obesity, 40.4% tobacco use, 40.3% HTN, 31.6% disability, 31.4% depression, 30.4% drug abuse (adults); 31.3% obesity (children)
				- Nutrition education (16 ACOs)
				- MTMs (14 ACOs)
				- Connection to other nutrition resources (14 ACOs)
				- PRx (10 ACOs)
				- Kitchen supplies (7 ACOs)
				- Food boxes or groceries (6 ACOs)
				- Home-delivered meals (4 ACOs)
					Health care utilization
					- Adults: Nutrition-related services associated with significant decreases in inpatient hospitalizations and ED visits, but not primary care visits
					  - Adults with enrollments > 90 days: Nutrition-related services associated with significant decreases in inpatient hospitalizations and ED visits, but not primary care visits
					- Children (overall and those with enrollments > 90 days): Nutrition-related services associated with decreases in all metrics, though findings did not reach statistical significance
					Health care costs
					- Adults (overall and those with enrollments > 90 days): Nutrition-related services associated with decreases in health care costs, though findings did not reach statistical significance
					- Children (overall and those with enrollments > 90 days): Nutrition-related services associated with increases in health care costs, though findings did not reach statistical significance

ACER = average cost-effectiveness ratio, ACO = Accountable Care Organization, ADLs = activities of daily living, AIDS/HIV = acquired immunodeficiency syndrome/human immunodeficiency virus, aIRR = adjusted incidence rate ratio, BRFSS = Behavioral Risk Factor Surveillance System, CA = California, CCA = Commonwealth Care Alliance, CHASS = Community Health and Social Services, CHD = coronary heart disease, CHF = congestive heart failure, CI = confidence interval, CMS = Centers for Medicare and Medicaid Services, COPD = chronic obstructive pulmonary disease, CVD = cardiovascular disease, DM = diabetes mellitus, DOC-M = diabetes obesity cardiovascular disease microsimulation, ED = Emergency Department, EBT = electronic benefit transfer, ESRD = end-stage renal disease, FFS = fee-for-service, FL = Florida, FQHC = Federally Qualified Health Center, GA = Georgia, HbA1c = hemoglobin A1c, HF = heart failure, HTN = hypertension, IADLs = instrumental activities of daily living, ICER = incremental cost-effectiveness ratio, ID = Idaho, IRR = incidence rate ratio, LOS = length of stay, MA = Massachusetts, MANNA = Metropolitan Area Neighborhood Nutrition Alliance, ME = Maine, MEPS = Medical Expenditure Panel Survey, MI (location) = Michigan, MI (health) = myocardial infarction, MMC = Maine Medical Center, MNT = medical nutrition therapy, mo = month, MOW = Meals on Wheels, MTM = medically tailored meal, NC = North Carolina, ND = North Dakota, NH = New Hampshire, NHANES = National Health and Nutrition Examination Survey, OH = Ohio, OK = Oklahoma, PA = Pennsylvania, PRx = produce prescription, QALY = quality adjusted life year, RD = Registered Dietitian, RI = Rhode Island, SNF = skilled nursing facility, TX = Texas, UI = uncertainty interval, US = United States, wk = week, WTP = willingness-to-pay, WV = West Virginia, yr = year

Discussion

Key Themes Surrounding Cost Effectiveness, Study Populations, and Study Settings

Though ICERs varied widely by CUA study, one simulation study of PRx projected a lifetime ICER of $18,100/QALY,[3] and one simulation study of a healthy incentive projected a lifetime ICER of $13,194–$18,184/QALY (depending on which food were covered by the incentive).[31] In comparison, when used as treatment for hypertension, the ICER for statins is $20,000/QALY, and when used as primary prevention, this increases to $37,000/QALY.[3] Though it is true that statins have more immediate effects than dietary changes,[44] they are most effective when combined with dietary change, and in some instances,[45] making dietary changes can allow patients to get off their statin.[46] Considering that common WTP thresholds are set at $150,000/QALY and $50,000/QALY[3,33] certain FIM approaches may be similar to, and in some cases even eliminate the need for, current medical treatment.

As mentioned above, the majority of studies were conducted in elderly or near-elderly populations. While individuals of all ages can benefit from a broader FAM approach, and while specific FIM interventions could theoretically help prevent disease-related conditions before they develop, currently FIM is most frequently used in older adults, as this population is most likely to have developed a diet-related condition and face increased health care costs.[47]

An additional key theme from the literature review is that the majority of FIM interventions are being spearheaded by private health-care payers, nonprofit organizations, and community-based organizations.[3,42] At a federal level, aside from the Gus Schumacher Nutrition Incentive Program[48] and medical nutrition therapy for patients with diabetes or kidney disease,[49] there has been limited coverage of FIM interventions. Such interventions typically come in the form of temporarily funded home health care benefits, Medicare Advantage programs, or state

Section 1115 and 1915 waivers—[3,17,21]though these approaches are not without their limitations.[32] This is despite the fact that both state and federal governments have experience conducting similar programs. For example, Meals on Wheels, the largest program funded by the Older Americans Act, provides millions of home-delivered meals to hundreds of thousands of low-income individuals above age 60.[37] States governments, especially those in California, Massachusetts, and North Carolina, have also begun accumulating expertise in rolling out FIM interventions.[17] Given the findings of this literature review, it is in both states' and the federal government's best interest to broaden FIM efforts.

Upfront Costs vs. Cost Effectiveness and Application to Policy Practice

Unfortunately, despite the evidence for FIM interventions, it is important to remember that cost effectiveness does not automatically cancel out upfront costs. Additionally, it is often the case that those paying for prevention efforts may not realize direct economic benefits from such efforts. In this review, PRx food costs ranged from \$42–\$47 per person per month—though amounts actually redeemed ranged from \$32–\$35.70—[3,31] and administrative costs were estimated to run two to three times higher than those of the Supplemental Nutrition Assistance Program.[3] Thus, regardless of the cost savings such programs can generate, they still require significant upfront funding, and money must first be spent before savings can be realized. At the end of the day, state governments must balance budgets, and the federal government has an obligation to taxpayers. As some authors put it, "determining where the initial investment to expand meal delivery would come from is not a small consideration at a time when policy makers are looking for ways to cut existing federal and state programs."[37]

However, the same argument can be made for current health care spending. In 2024, the US spent \$14,885 per capita on health care compared to its peers' average of \$7,371.[9] Health care costs consumed 17.2% of gross domestic product compared to 11.2% of peer nations,[50] and for all this spending, the US sees some of the worst health outcomes, with a life expectancy 4.1 yr shorter than that of peer nations in 2023.[51] While it may be challenging to find the upfront costs needed to fund FIM interventions, current spending is certainly not sustainable. When individuals are inadequately nourished for extended periods of time, these costs show up in areas such as emergency room visits, inpatient hospital stays, prescription drug costs, dental visits, home care, and lost productivity.[3]

In light of these challenges, rolling out FIM interventions in a multi-staged approach presents an opportunity for compromise. For example, in this literature review, a simulation study providing 10 MTMs per week for 8 months to 10.4 million Americans at a cost of $11.15 per meal resulted in projected cost savings in 49 of 50 states, with an average cost savings of $2,287 per patient.[32] However, when restricted to only the 3.95 million patients with diabetes, projected cost savings were achieved in all 50 states with an average cost savings of $3,403 per patient.[32] When further restricted to only the 1.79 million patients with heart failure, projected cost savings were again achieved in all 50 states with an average cost savings of $3,989 per patient.[32] From this lens, states can roll out FIM programs first in populations that stand the most to benefit—such as those on Medicare[31] with diabetes, heart failure, food insecurity—and, as cost savings are realized, further expand programs, creating a positive cycle of health and economic benefit.

At the same time, it is important to remember an important goal of FIM, which is to provide high-quality care for individuals with diet-related conditions to improve their health.[32] Though cost analyses are necessary to make coverage and policy decisions, viewing them solely as cost containment strategies creates a "high bar that is unrealistic for most therapies, preventive services, and diagnostic testing used in health care today."[32] Additionally, FIM interventions often come with additional benefits that cannot be captured in economic analyses. For example, participants in programs such as Meals on Wheels often report social benefits from interacting with those who prepare and deliver their meals.[37] Further benefits are also seen for caregivers' well-being and health equity.[32]

Strengths and Limitations

The strengths of this review include a thorough comparison of the different types of economic evaluations and FIM interventions. This review also excluded any study that did not conduct a form of economic analysis. Therefore, unlike other reviews that compile both health outcome and cost findings, this review allows for a focused, in-depth look at economic evaluations of FIM interventions. Additionally, this review utilized a systematic approach and provided detailed information about how searches were conducted, allowing for replicability.

The limitations of this review include the fact that, unlike in other reviews where disagreement between reviewers can lead to further refinement of the search strategy, there was one reviewer. Multiple authors could also have led to a larger number of articles captured through snowballing. Similarly, as with all reviews, there is the potential that the search strategy did not capture all applicable studies, particularly unpublished and grey literature. Additional limitations include those inherent in each study design (e.g., simulation studies only reflect the most likely scenarios based on the best available data). Additionally, because the majority of studies evaluated older adults with high health needs, findings cannot be generalized to younger or healthier populations.

Areas for Future Work

This literature review summarized current evidence on economic evaluations of FIM interventions and revealed valuable areas for future research. For example, simulation studies primarily made up the most recent and extensive research. While valuable, simulation studies have significant limitations. Future FIM efforts should include rigorous evaluation of both economic and health care data to allow for examination of real-world data. Additionally, this review covered a wide variety of economic evaluations, limiting the ability to make effective comparisons. Where possible, future research should prioritize rigorous forms of economic evaluations, preferably those that can be compared across interventions, such as CUA. Future research should also explicitly consider and discuss feasibility, as only one study systematically evaluated this.

Furthermore, the majority of studies evaluated MTMs. While valuable, this FIM intervention may face more implementation barriers compared to PRx due to higher costs.[6] Therefore, future research should bolster economic evaluations of less researched, lower cost, and potentially more feasible interventions such as PRx and incentive programs. Additionally, the original goal of this literature review was to explore economic evaluations of FIM interventions offered through government mechanisms, such as Section 1115 and 1915 waivers. However, due to the significant lack of peer-reviewed, published literature evaluating the cost effectiveness of these programs, this goal was abandoned and the search strategy broadened. Therefore, future research should conduct economic evaluations on FIM interventions offered through government payers.

Conclusion

Though FIM is a relatively new concept, its evidence base has been steadily growing over the past few decades. To date, the majority of evidence has focused on impacts on health outcomes, with a minority evaluating its ability to improve health outcomes while reducing health care costs. This review provides an introduction to FIM and economic evaluation and examines existing economic evidence for FIM. Despite heterogeneity in study population, FIM approach, and economic evaluation method, FIM was consistently associated with improved health outcomes and reduced health care utilization, findings which often translated into cost savings. Observational studies frequently demonstrated immediate reductions in health care utilization and costs, while simulation studies suggested FIM approaches may fall within commonly accepted WTP thresholds, particularly over longer time horizons.

Despite these findings, important challenges remain. The wide variation in study populations, FIM approaches, and economic evaluation methods limited the ability to effectively compare results across studies. Additionally, much of the strongest economic evidence relies on simulation rather than real-world implementation, and the majority of economic evaluations are driven by private insurers, nonprofit organizations, and community-based entities. The latter suggest a need for increased government involvement to support broader and more equitable implementation. Furthermore, although FIM interventions may be cost effective, they still require substantial upfront investment, raising practical and political considerations for large-scale adoption.

Nevertheless, this review suggests that FIM is a promising strategy to address the interconnected challenges of chronic diet-related disease, food insecurity, barriers to healthy food access, and rising health care costs. Policymakers, health care systems, and researchers may benefit from prioritizing targeted, phased FIM approaches that focus on high-need populations, while also continuing to build economic evidence using rigorous, standardized economic evaluation methods. Ultimately, expanding FIM efforts has the potential not only to improve population health and reduce health care costs, but also to build a more sustainable and equitable health care system.

References

1. Ridberg R, Sharib JR, Garfield K, Hanson E, Mozaffarian D. 'Food is Medicine' in the US: A national survey of public perceptions of care, practices, and policies. *Health Aff (Millwood)*. 2025;44(4). Accessed March 9, 2026. https://www.healthaffairs.org/doi/10.1377/hlthaff.2024.00585

2. Health I of M (US) C on D and, Woteki CE, Thomas PR. Diet and chronic disease in the United States. In: *Eat for Life: The Food and Nutrition Board's Guide to Reducing Your Risk of Chronic Disease*. National Academies Press (US); 1992. Accessed March 9, 2026. https://www.ncbi.nlm.nih.gov/books/NBK235021/

3. Wang L, Lauren BN, Hager K, et al. Health and economic impacts of implementing produce prescription programs for diabetes in the United States: A microsimulation study. *J Am Heart Assoc*. 2023;12(15):e029215. doi:10.1161/JAHA.122.029215

4. Kim DD, Wang L, Lauren BN, et al. Development and validation of the US Diabetes, Obesity, Cardiovascular Disease Microsimulation (DOC-M) model: Health disparity and economic impact model. *Med Decis Mak Int J Soc Med Decis Mak*. 2023;43(7-8):930-948. doi:10.1177/0272989X231196916

5. U.S. Department of Health and Human Services. Food insecurity. Healthy People 2030. https://odphp.health.gov/healthypeople/priority-areas/social-determinants-health/literature-summaries/food-insecurity

6. Berkowitz SA, Terranova J, Randall L, Cranston K, Waters DB, Hsu J. Association between receipt of a medically tailored meal program and health care use. *JAMA Intern Med*. 2019;179(6):786-793. doi:10.1001/jamainternmed.2019.0198

7. Gregory CA, Reed-Jones M. Chronic disease risk increased with U.S. household food insecurity. Charts of Note. April 4, 2024. Accessed March 9, 2026. https://www.ers.usda.gov/data-products/charts-of-note/chart-detail?chartId=108913

8. Berkowitz SA, Basu S, Meigs JB, Seligman HK. Food insecurity and health care expenditures in the United States, 2011–2013. *Health Serv Res*. 2018;53(3):1600-1620. doi:10.1111/1475-6773.12730

9. Sparks G, Lopes L, Montero A, Presiado M, Hamel L. Americans' challenges with health care costs. KFF. January 29, 2026. Accessed March 9, 2026. https://www.kff.org/health-costs/americans-challenges-with-health-care-costs/

10. Radley DC, Shah A, Collins SR, Powe NR, Zephyrin LC. *Advancing Racial Equity in U.S. Health Care: The Commonwealth Fund 2024 State Health Disparities Report*. The Commonwealth Fund; 2024. doi:10.26099/vw02-fa96

11. Fast facts: Health and economic costs of chronic conditions. Centers for Disease Control and Prevention. August 14, 2025. Accessed March 19, 2026. https://www.cdc.gov/chronic-disease/data-research/facts-stats/index.html

12. Racial and ethnic health care disparities. Center for Medicare Advocacy. Accessed March 19, 2026. https://medicareadvocacy.org/medicare-info/health-care-disparities/

13. Garcia SP, Haddix A, Barnett K. Incremental health care costs associated with food insecurity and chronic conditions among older adults. *Prev Chronic Dis*. 2018;15:E108. doi:10.5888/pcd15.180058

14. Claude. Version April 12. Claude; 2026. Accessed April 12, 2026. https://claude.ai.

15. Healthy food environments. CDC Nutrition. January 23, 2025. Accessed April 12, 2026. https://www.cdc.gov/nutrition/php/healthy-food-environments/index.html

16. Flatt AM. *Food Is Medicine: Advancing Health Equity through Nutrition*. American Public Health Association; 2025. https://www.apha.org/getcontentasset/53b9b8f1-3df4-4a6a-b65d-46fc0f7de0d9/7ca0dc9d-611d-46e2-9fd3-26a4c03ddcbb/food-is-medicine-report.pdf?language=en

17. Hager K, Cudhea FP, Wong JB, et al. Association of national expansion of insurance coverage of medically tailored meals with estimated hospitalizations and health care expenditures in the US. *JAMA Netw Open*. 2022;5(10):e2236898. doi:10.1001/jamanetworkopen.2022.36898

18. Yoder AD, Proaño GV, Handu D. Retail nutrition programs and outcomes: An evidence analysis center scoping review. *J Acad Nutr Diet*. 2021;121(9):1866-1880.e4. doi:10.1016/j.jand.2020.08.080

19. Volpp KG, Berkowitz SA, Sharma SV, et al. Food Is Medicine: A presidential advisory from the American Heart Association. *Circulation*. 2023;148(18):1417-1439. doi:10.1161/CIR.0000000000001182

20. Hanson C, Conklin K. "Food Is Medicine" vs "Food As Medicine": Why the distinction matters for local food systems. New Venture Advisors LLC. June 19, 2025. Accessed March 12, 2026. https://www.newventureadvisors.net/food-is-medicine-vs-food-as-medicine/

21. *Advances in the Food Is Medicine Field: Annual Report 2025*. American Heart Association; 2025. https://www.healthcarexfood.org/en/-/media/Files/HCxF/AdvancesintheFieldofFIM_AnnualReport2025.pdf?sc_lang=en

22. Office of Disease Prevention and Health Promotion. Types of interventions. Food is Medicine. September 12, 2024. Accessed March 9, 2026. https://odphp.health.gov/foodismedicine/understanding-food-medicine/types-interventions

23. Mozaffarian D, Liu J, Sy S, et al. Cost-effectiveness of financial incentives and disincentives for improving food purchases and health through the US Supplemental Nutrition Assistance Program (SNAP): A microsimulation study. *PLoS Med*. 2018;15(10):e1002661. doi:10.1371/journal.pmed.1002661

24. Nutrition Assessment, Counseling, and Support (NACS): A user's guide -- Module 3: Nutrition education and counseling, version 2. Published online 2016. Accessed April 12, 2026. https://www.fantaproject.org/sites/default/files/resources/NACS-Module-3-Counseling-May2016.pdf

25. Science of Health Care by Food. Health Care x Food. Accessed March 19, 2026. https://www.healthcarexfood.org/en/science-of-hcxf

26. Welfare economics. EBSCO. Accessed March 19, 2026. https://www.ebsco.com

27. Kim Y, Kim Y, Lee HJ, et al. The primary process and key concepts of economic evaluation in healthcare. *J Prev Med Pub Health*. 2022;55(5):415-423. doi:10.3961/jpmph.22.195

28. Boyer NRS. *Economic evaluation of population health interventions aimed at children and delivered at school*. University of Glasgow; 2017. https://theses.gla.ac.uk/9012/7/2018BoyerPhD.pdf

29. Sin CH, McMahon A. Making the economic case for nursing innovations: Horses to courses. *Nurs Manag (Harrow)*. 2014;21(4). https://rcni.com/sites/rcn_nspace/files/Economic-Assessment-2015.pdf

30. Baumgartner JN, Headley J, Swanson K, Koirala N, Datta S. Economic impact: Finding the right analysis for your evaluation needs. Published online 2017. Accessed April 12, 2026. https://globalhealth.duke.edu/sites/default/files/files/el_econtool.pdf

31. Wang L, Cudhea F, Deng S, et al. Health and economic impact and cost-effectiveness of produce prescriptions for diabetes in 50 U.S. states: A microsimulation study. *Diabetes Care*. 2025;48(10):1783-1793. doi:10.2337/dc25-0144

32. Deng S, Hager K, Wang L, et al. Estimated impact of medically tailored meals on health care use and expenditures in 50 US states. *Health Aff Proj Hope*. 2025;44(4):433-442. doi:10.1377/hlthaff.2024.01307

33. Lee Y, Mozaffarian D, Sy S, et al. Cost-effectiveness of financial incentives for improving diet and health through Medicare and Medicaid: A microsimulation study. *PLoS Med*. 2019;16(3):1-20. doi:10.1371/journal.pmed.1002761

34. Pandya A, Sy S, Cho S, Alam S, Weinstein MC, Gaziano TA. Validation of a cardiovascular disease policy micro-simulation model using both survival and receiver operating characteristic curves. *Med Decis Mak Int J Soc Med Decis Mak*. 2017;37(7):802-814. doi:10.1177/0272989X17706081

35. Hayes K, Serafini M, Hoagland WG, et al. *Next steps in chronic care: Expanding innovative Medicare benefits*. Bipartisan Policy Center; 2019. Accessed March 8, 2026. https://bipartisanpolicy.org/wp-content/uploads/2019/07/Next-Steps-in-Chronic-Care.pdf"

36. Raso D. *Cost-effectiveness of a produce prescription intervention*. Yale School of Public Health; 2020. https://www-proquest-com.ezp.slu.edu/docview/2415816810?pq-origsite=wos&accountid=8065&sourcetype=Dissertations%20&%20Theses

37. Thomas KS, Mor V. Providing more home-delivered meals is one way to keep older adults with low care needs out of nursing homes. *Health Aff Proj Hope*. 2013;32(10):1796-1802. doi:10.1377/hlthaff.2013.0390

38. Berkowitz SA, Terranova J, Hill C, et al. Meal delivery programs reduce the use of costly health care in dually eligible Medicare and Medicaid beneficiaries. *Health Aff (Millwood)*. 2018;37(4):535-542. doi:10.1377/hlthaff.2017.0999

39. Gurvey J, Rand K, Daugherty S, Dinger C, Schmeling J, Laverty N. Examining health care costs among MANNA clients and a comparison group. *J Prim Care Community Health*. 2013;4(4):311-317. doi:10.1177/2150131913490737

40. Martin SL, Connelly N, Parsons C, Blackstone K. Simply delivered meals: A tale of collaboration. *Am J Manag Care*. 2018;24(6):301-304.

41. Shan M, Gutman R, Dosa D, et al. *More than a meal: Medicare claims analysis*. Meals on Wheels America; 2017. Accessed March 5, 2026. https://member.mealsonwheelsamerica.org/wp-content/uploads/2025/04/Meals-on-Wheels-America-2017.pdf

42. Haddad EN, Miles R, Alejandro-Rodriguez M, et al. Feasibility of self-investment in a medically tailored meals program by a large health enterprise: Cleveland Clinic experience. *Nutr Health*. 2025;31(4):1503-1513. doi:10.1177/02601060241307980

43. Hager K, Sabatino M, Williams J, et al. Medicaid nutrition supports associated with reductions in hospitalizations and ED visits in Massachusetts, 2020-23. *Health Aff (Millwood)*. 2025;44(4):413-421. doi:10.1377/hlthaff.2024.01409

44. Gandjour A. Health-economic evaluation of statins versus lifestyle changes for cardiovascular disease prevention. *PLOS One*. 2025;20(9):e0331176. doi:10.1371/journal.pone.0331176

45. Rathner JL. Statin medications are not a blank check for eating poorly. Laborer’s Health & Safety Fund of North America. July 16, 2014. Accessed March 20, 2026. https://lhsfna.org/statin-medications-are-not-a-blank-check-for-eating-poorly/

46. Vann M. Stuck with statins? Try a plant-based diet to cut back. MedShadow Foundation. September 16, 2020. Accessed March 20, 2026. https://medshadow.org/integrative-health/nutrition-lifestyle/stuck-with-statins-try-a-plant-based-diet-to-cut-back/

47. O'Neill Hayes T, Asres R. *The Economic Costs of Poor Nutrition*. American Action Forum; 2022. Accessed March 20, 2026. https://www.americanactionforum.org/research/the-economic-costs-of-poor-nutrition/

48. Gus Schumacher Nutrition Incentive Program. USDA Food and Nutrition Service. September 17, 2025. Accessed March 20, 2026. https://www.fns.usda.gov/snap/gusnip-grant-program

49. Medical nutrition therapy benefit for diabetes & ESRD. Centers for Medicare & Medicaid Services. Accessed March 20, 2026. https://www.cms.gov/medicare-coverage-database/view/ncacal-decision-memo.aspx?proposed=N&NCAId=53

50. Telesford I, Cotter L, Wager E, Cox C. How does health spending in the U.S. compare to other countries? Peterson-KFF Health System Tracker. March 11, 2026. Accessed March 20, 2026. https://www.healthsystemtracker.org/chart-collection/health-spending-u-s-compare-countries/

51. Telesford I, Wager E, Cox C. How does the quality of the U.S. health system compare to other countries? Peterson-KFF Health System Tracker. October 6, 2025. Accessed March 20, 2026. https://www.healthsystemtracker.org/chart-collection/quality-u-s-healthcare-system-compare-countries/

Appendix

Table A1: Search Strategy

Search String	("cost minimization" OR "cost-minimization" OR "cost benefit" OR "cost-benefit" OR "cost effectiveness" OR "cost-effectiveness" OR "cost utility" OR "cost-utility" OR "cost offset" OR "cost-offset" OR "cost consequence" OR "cost-consequence" OR "cost analysis" OR "cost savings" OR "economic analysis" OR "economic evaluation" OR "budget impact" OR "budgetary impact" OR "return on investment" OR "ROI") AND ("medically tailored meals" OR "medically tailored groceries" OR "medically supportive groceries" OR "produce prescription" OR "produce Rx" OR "fruit and vegetable prescription" OR "food as medicine" OR "food is medicine" OR "food pharmacy" OR "home delivered meals" OR "home-delivered meals")
Databases	PubMed, Web of Science, Cumulative Index to Nursing and Allied Health Literature (CINAHL), EconLit, Agricola, and Business Source Premier

Because there is limited data on cost effectiveness of FIM interventions, and because economic analyses are described in a variety of ways, the economic portion of the search string was kept intentionally broad. However, when designing the FIM portion of the string, the initial list of terms captured a large amount of results, many of which were not relevant to the goals of the review. After reviewing results, the review was narrowed to FIM interventions that provided food, excluding originally included terms of "nutrition counseling," "nutrition instruction," and "nutrition education." The search string was further condensed by individually searching each original search term and excluding terms that resulted in either zero results or results with limited relevance to the goals of the review. These terms included the following: "medically tailored food," "tailored meals," "produce incentive," "healthy food incentive," "food benefit," "food intervention," "medically supportive meals," "nutrition incentive," "pantry stocking," "nutrition prescriptions," and "grocery provision."

Table A2: Detailed results of literature review

Authors, Year	Study Design, Type of Economic Evaluation	Study Location, Year	Study Population	Intervention	Results
Simulation Studies					

Wang et al., 2023[3]	Microsimulation modeling study, CUA	US, +5, +10, and +25 yr from analysis	6.5 million US adults using data from NHANES 2013–2018, augmented with data from 244.4 million adults from MEPS 2014–2016 (DOC-M model)[4]
			- Mean age: 58.2 ± 10.2 yr
			- Sex: 55.6% female
			- Race & ethnicity: 43.1% non-Hispanic White, 17.2% non‑Hispanic Black, 29.0% Hispanic, 10.7% other
			- Education: 37.1% less than high school, 26.5% high school, 36.4% college
			- Family income to poverty ratio: 56.6% <1.30, 17.1% 1.30–1.84, 16.4% 1.85–2.99, 9.9% >3.0
			- Insurance: 29.4% private insurance, 26.9% Medicare, 26.9% Medicaid, 12.5% dual
				Produce prescription at $42/mo per food box or voucher ($32/mo actually redeemed)	Over 25 yr (lifetime)
					- Prevent 292,000 CVD events (95% UI 143,000–440,000)
					  - Effects stronger in younger (<65 yr) adults; non-Hispanic Black and Hispanic adults; and uninsured, Medicaid, or privately insured adults
					- Generate 260,000 QALYs (95% UI 110,000–411,000)
					- Cost $37.3 billion in food costs (95% UI $27.0 billion–$47.5 billion) and $6.99 billion in administrative costs (95% UI $5.07 billion–$8.19 billion)
					- Save $39.6 billion in health care costs (95% UI $20.5 billion–$58.6 billion) and $4.8 billion in productivity costs (95% UI $1.84 billion–$7.70 billion)
					  - Health care cost findings not robust in sensitivity analysis, ranging from $24.1 billion in additional costs at the 2.5th percentile of intervention effects to $15.3 billion in savings at the 97.5th percentile
					- Net societal cost savings of $50 million
					- ICER = $18,100/QALY
					  - ICERs similar across all population subgroups
					  - Health care and societal cost effectiveness robust in sensitivity analysis, persisting when administrative costs varied from 8–23% of total program costs (health care ICERs ranging from $9,300/QALY–$36,800/QALY; societal ranging from cost-saving–$18,500/QALY)
					  - Cost effectiveness not robust in sensitivity analysis, persisting with 50%

			eligible, 16.5% no insurance
			- Health: 100.0% DM, 74.8% HTN, 30.0% CVD, 100.0% food insecurity, BMI 33.6 ± 7.95, HbA1c 7.3 ± 1.95
			- Diet: fruit consumption 0.86 ± 1.08 servings/day, vegetable consumption 1.30 ± 1.04 servings/day
					participation (net health care costs of $5.86 billion and ICER of $450,000/QALY and societal costs of $3.47 billion and ICER of $26,7000/QALY)
					- Cost effectiveness persisted until food + administrative costs exceeded $67.3/mo per box or voucher
					- 98.4% probability of net health care cost effectiveness with WTP of $150,000, and 74.4% with WTP of $50,000
					- 98.9% probability of societal cost effectiveness with WTP of $150,000, and 84.6% with WTP of $50,000
					- Health care and societal cost effectiveness robust in sensitivity analysis, ranging from $102,000/QALY–cost-saving across across the 2.5th–97.5th percentiles of intervention effects
					
					Over 10 yr
					- Prevent 126,000 CVD events (95% UI 62,300–190,000)
					  - Effects stronger in older (>65 yr) and non-Hispanic Black and Hispanic adults
					- Health care ICER = $58,000/QALY and societal ICER = $28,700/QALY
					  - Effects stronger in older (>65 yr) adults and adults with Medicare or Medicaid
					
					Over 5 yr
					- Prevent 66,900 CVD events (95% UI 31,900–102,000)
					  - Effects stronger in older (>65 yr) and non-Hispanic Black and Hispanic adults
					- Health care ICER = $92,700/QALY and societal ICER = $66,100/QALY
					  - Effects stronger in older (>65 yr) adults and adults with Medicare or Medicaid

Wang et al., 2025[31]	Microsimulation modeling study, CUA	US, +5 and +10 yr from analysis	5,555,000 US adults using data from NHANES 2011–2020 and BRFSS 2021–2023, augmented with data from the MEPS 2017–2022 and 2023 Medicare Inpatient Prospective Payment data (DOC-M model)[4]
			- Mean age: ranged from 57–62 yr depending on the state
			- Sex: ranged from 36%–59% depending on the state
			- Race & ethnicity: non-Hispanic White ranged from 12.5%–91.8%, Black from 0.4%–46.9%, Hispanic from 0.7%–68.9% depending on the state
			- Insurance: Private insurance ranged from 23.2%–43.8%,
				Produce prescription at $47/mo per food box or voucher ($35.70 /mo actually redeemed)	Over 10 yr
					- Prevent minimum of 94 CVD events in WY (95% UI 48–147) to maximum of 9,240 CVD events in TX (95% UI 4,710–14,500)
					  - When standardized per 10,000 patients, prevented minimum of 137 CVD events in ID (95% UI 69–214) to maximum of 181 (95% UI 89–385) in FL
					  - Effects stronger in adults on Medicare (net cost-saving in 48/50 states), Medicaid (net cost-saving in 41/50 states), and private payers (net cost-saving in 29/50 states)
					- Generate minimum of 92.2 QALYs in AK (95% UI 30.8–159) to maximum of 9,990 QALYs in CA (95% UI 4,810–15,500)
					  - When standardized per 10,000 patients, generated minimum of 138 QALYsin WV (95% UI 60.9–223) to 166 in NH (95% UI 68.6–279)
					- Implementation costs ranged from minimum of $25.2 million in AK (95% UI $18.5 million–$32.3 million) to maximum of $2.74 billion in CA (95% UI $2.01 billion–$3.35 billion), with nationwide costs of $26.1 billion
					- Health care cost savings ranged from minimum of $25.3 million in AK (95% UI $11 million–$44.2 million) to maximum of $2.59 billion in CA (95% UI $1.12 billion–$4.33 billion), with nationwide savings of $29.2 billion
					  - When standardized per 10,000 patients, net health care costs ranged from $10.2

			Medicaid from 7.7%–32.1%, and Medicare from 20.0%–47.9% depending on the state
			- Health: BMI ranged from 32–36 kg/m^2 and history of CVD ranged from 19.3%–37.8% depending on the state, 100.0% DM, 100.0% food insecurity
					million in HI to -\$11.8 million in ME
					- ICER <\$50,000/QALY in 49/50 states (\$66,500/QALY in HI)
					  - ICER <\$50,000/QALY in 50/50 states for Medicare, 48/50 states for Medicaid, and 48/50 states for private payers
					  - In sensitivity analyses, cost savings achieved in 34/50 states, high cost effectiveness (\$50,000/QALY) in 14/50 states, and cost effectiveness (\$150,000/QALY) in 2/50 states
					- Health care net savings projected in 43/50 states, ranging from minimum of \$155 million net costs in CA (95% UI -\$1,790 million–\$1,830 million) to maximum of \$297 million net savings in NY (95% UI -\$669 million–\$1,490 million), with nationwide health care net savings of \$2.68 billion
					  - When standardized per 10,000 patients, averted health care costs ranged from minimum of \$38.5 million in OK (95% UI 16.8–65.6) to maximum of \$50.6 million in MA (95% UI 22.0–87.9)
					- Societal net savings projected in 48/50 states and cost effectiveness in 49/50 states (\$53,600/QALY in HI), with nationwide societal net savings of \$4.17 billion
					  - When standardized per 10,000 patients, net societal costs ranged from minimum of \$8.23 million in HI to -\$14.1 million in ME
					- Findings remained robust when excluding 2021 BRFSS (COVID-19) data
					
					Over 5 yr
					- Similar patterns of health benefits and economic outcomes were observed, with cost savings projected in 40/50 states

Hager et al., 2022[17]	Cohort simulation modeling study, CEA	US, +1 and +10 yr from analysis	6,309,998 US adults using data from the MEPS 2010–2019
			- Mean age: 68.1 ± 16.6 yr
			- Sex: 63.4% female, 36.6% male
			- Race & ethnicity: 66.7% non-Hispanic White, 14.2% non-Hispanic Black, 11.3% Hispanic, 3.1% non-Hispanic Asian, 4.7% other
			- Median ratio of household income to poverty level: 2.8 ± 2.7
			- Insurance: 40.7% Medicare, 11.1% Medicaid, 24.7% dual eligible, 23.5% private payer
			- Health: 100.0% with limitations in IADLs, 44.9% with DM, 37.2% with cancer, 36.5% with other heart disease,
				Providing 10 MTMs/wk for 8 months/yr at a cost of \$9.30 per meal	Over 10 yr
					- Prevent 18,257,000 hospitalizations (95% UI 14,690,000–22,109,000)
					- Program costs of \$298.7 billion (95% UI \$279.7 billion–\$317.4 billion)
					- Reduce health care expenditures by \$484.5 billion (95% UI \$310.2 billion–\$678.4 billion)
					- Net health care savings of \$185.1 billion (95% UI \$12.9 billion–\$377.8 billion)
					- Findings persisted in sensitivity analyses
					
					Over one year
					- Prevent 1,594,000 hospitalizations (95% UI 1,297,000–1,912,000)
					- Program costs of \$24.8 billion (95% UI of \$23.1 billion–\$26.8 billion)
					  - Per-meal cost would need to increase from \$9.30 to \$18.89 for MTMs to become cost neutral instead of cost saving
					- Reduce health care expenditures by \$38.7 billion (95% UI \$24.9 billion–\$53.9 billion)
					- Net health care savings of \$13.6 billion (95% UI of \$0.2 billion–\$28.5 billion)
					  - \$5.9 billion (95% UI −\$1.9 billion–\$14.1 billion) for dual-eligible individuals, \$3.4 billion (95% UI −\$5.4 billion–\$12.1 billion) for Medicare, \$3.1 billion (95% UI −\$2.9 billion–\$9.5 billion) for private payers, and \$1.7 billion (95% UI −\$1.1 billion–\$5.1 billion) for Medicaid

			36.1% with stroke, 26.8% with CHF, 21.3% with MI - Health care use: average number of annual ED emissions 0.98 ± 1.68, average number of annual hospitalizations 0.54 ± 0.94, average annual health care expenditures $31,134 ± $34,749,		- Findings persisted in sensitivity analyses

Deng et al., 2025[32]	Cohort simulation modeling study, CEA	US, +1 and +5 yr from analysis	10,358,159 US adults using data from the MEPS 2010–2019, augmented with data from the 2019 BRFSS
			- Mean age: 67.8 ± 16.5 yr
			- Sex: 62.7% female, 34.3% male
			- Race & ethnicity: 64.4% non-Hispanic White, 15.9% non-Hispanic Black, 12.8% Hispanic, 6.8% other or multiple
			- Education: 67.3% received Medicare or Medicaid
			- Mean family income as a percent of the FPL: 267.9%
			- Insurance: 55.5% Medicare, 11.8% Medicaid, 26.3% dual eligible, 30.3% private payer
			- Health: 74.9% HTN, 38.4% DM, 33.8% lung disease, 26.9% CHD, 21.9% cancer, 20.0% stroke, 100.0% 1+ IADL limitations
				Providing 10 MTMs/wk for 8 months/yr at a cost of $11.15 per meal	Over five yr
					- Prevent 10,792,000 hospitalizations nationwide (95% UI 7,298,000–14,333,000)
					  - Range from a minimum of 11,000 in WY (95% UI -33,000–58,000) to maximum of 1,091,000 in NC (95% UI 686,000–1,524,000)
					- Net policy cost savings of $111.1 billion nationwide (95% UI -$44.3 billion–$282.0 billion)
					  - Net policy cost savings findings not robust in sensitivity analyses
					
					Over one year
					- Prevent 2,607,200 hospitalizations nationwide (95% UI 1,675,700–3,558,100)
					  - Range from minimum of 2,400 in WY (95% UI -6,800–12,400) to maximum of 211,200 in FL (95% UI 133,900–294,600)
					  - On average, 4.0 patients nationwide would need to receive MTMs to avert one hospitalization (95% UI 2.9–6.2)
					    - Ranged from minimum of 2.3 in MD (95% UI 1.3–5.8) to 6.9 in CO (95% UI 4.3–13.8)
					  - Hospitalization findings robust in sensitivity analyses
					- Program costs of $37,346 million nationwide (95% UI $36,664 million–$37,990 million)
					  - Ranged from minimum of $46 million in ND (95% UI $45 million–$47 million) to maximum of $3,329 million in CA (95% UI $3,273 million–$3,390 million)
					  - Per-meal cost would need to increase from $11.15 to $18.30 to become cost neutral instead of cost saving
					- Health care cost savings of $61,040 million nationwide (95% UI $22,031 million–$102,932 million)
					  - Ranged from minimum of $65 million in ND (95% UI $22 million–$106 million) to maximum of $6,120 million in CA (95% UI $2,065 million–$10,430 million)
					- Net policy cost savings of $23,694 million nationwide (95% UI -$14,633 million–$64,942 million)
					  - Projected cost savings in 49/50 states, ranging from minimum of -$35 million in AL (95% UI -$593 million–$576 million) to maximum of $2,791 million in

					CA (95% UI -$1,209 million–$7,040 million) - Findings not robust in sensitivity analyses - Net per patient health care savings of $2,287 nationwide (95% UI -$1,413–$6,270) - Ranged from minimum of $3,248 per patient in MD (95% UI -$1,086–$8,352) to $6,299 per patient in CT (95% UI $15–$13,974)
Lee et al., 2019[33]	Microsimulation modeling study, CUA	US, +5, +10, +20, and +lifetime from analysis	82,000,000 US adults using data from NHANES 2009–2014 (CVD-PREDICT model)[34] - Mean age: 68.1 ± 11.4 yr - Sex: 56.2% female - Race & ethnicity: 74.5% non-Hispanic White, 11.8% non-Hispanic Black, 9.4% Hispanic, 9.4, 4.4% other - Insurance: 70.98% Medicare, 42.93% Medicaid (13.90% dual-eligible) - Education: 25.8% less than high school, 54.7% high school, 19.5% college - Poverty to income ratio: 35.0% <1.30, 65.0% ≥1.30 - SNAP status: 67.4% ineligible, 19.0% participants, 13.6% eligible non-participants - Health: systolic blood pressure 130.3 ± 19.7 mmHg, diastolic blood pressure 40.6 ± 35.1 mmHg, BMI 29.4 ± 6.7 kg/m^2, total cholesterol 194.4 ± 42.6 mg/dL, HDL cholesterol 54.6 ± 16.4 mg/dL, 14.7% current smoker, 57.3% current HTN treatment, 20.2% DM, 9.8% MI, 8.3% stroke, 5.9% angina - Diet: 130.3 ± 134.2 g fruit/day, 175.7 ± 133.2 g vegetable/day, 24.3 ± 28.0 g whole grains/day, 11.9 ± 30.1 g nuts/seeds/day, 20.6 ± 45.0 g seafood/day, 23.1 ± 13.0 g plant oils/day	30% subsidy on fruits and vegetables (FV), provided through EBT card (~ $110/ individual/ year) 30% subsidy on fruits, vegetables, whole grains, nuts, seeds, seafood, and plant oils (FV+), provided through EBT card (~ $185/ individual/ year)	Over a lifetime - FV incentive - Prevent 1.93 million CVD events (95% UI 1.57 million–2.31 million) and 0.35 million CVD deaths (95% UI 0.28 million–0.42 million) - Generate 4.64 million QALYs (95% UI 3.69 million–5.69 million) - Cost $7.11 billion in administrative costs (95% UI $4.98 billion–$9.81 billion) and $115.5 billion in food subsidy costs (95% UI $80.9 billion–$159.5 billion) - Health care savings of $39.7 billion (95% UI $31.8 billion–$48.7 billion) - Net costs of $83.5 billion (95% UI $45.2 billion–$129.0 billion) - Overall ICER of $18,184/QALY (95% UI $9,270/QALY–$29,371/QALY) - Medicare ICER of $17,842 (95% UI $9,392/QALY–$28,998/QALY) - Medicaid ICER of $16,933 (95% UI $8,295/QALY–$28,007/QALY) - Dual-eligible ICER of $17,238 (95% UI $7,995/QALY–$29,407/QALY) - Societal net costs of $68.8 billion - Societal ICER of $14,576/QALY - FV+ incentive - Prevent 3.28 million CVD events (95% UI 2.87 million–3.69 million), 0.62 million CVD deaths (95% UI 0.54 million–0.71 million), and 0.12 million DM cases (95% UI 0.10 million–0.15 million) - Generate 8.40 million QALYs (95% UI 7.23 million–9.58 million) - Cost $12.2 billion in administrative costs (95% UI $9.61 billion–$15.1 billion) and $198.2 billion in food subsidy costs (95% UI $156.3 billion– $245.4 billion) - Health care savings of $100.2 billion (95% UI $87.9 billion–$113.9 billion) - Net costs of $111.1 billion (95% UI $67.0 billion–$160.6 billion) - ICER of $13,194/QALY (95% UI $7,741/QALY–$19,683/QALY) - Ranged from $10,868/QALY in dual-eligible adults to $11,453/QALY in Medicaid adults to $13,203/QALY in Medicare adults - Societal costs of $80.5 billion - Societal ICER of $9,497/QALY - Ranged from $5,842/QALY in dual-eligible adults to $5,916/QALY in Medicaid adults to $10,078/QALY in Medicare adults - Health benefits increased with program duration in both FV and FV+ - Cost-effectiveness probability of 1.00 for overall, Medicaid, dual-eligible, and Medicare adults in both FV and FV+

					- Findings generally robust in sensitivity and subgroup analyses, with ICERs by insurance group, age, race, education, income, and SNAP status <$50,000/QALY
					
					Over 10 yr
					- FV+ incentive
					  - ICER of $35,246/QALY in dual-eligible adults, $34,721/QALY in Medicare adults, and $58,888/QALY in Medicaid adults
					
					Over five yr
					- FV incentive
					  - Cost-effectiveness probability 0.886 overall
					    - Ranged from 0.506 for Medicaid adults to 0.859 for dual-eligible adults to 0.909 for Medicare adults
					- FV+ incentive
					  - ICER of $77,537/QALY in dual-eligible adults, $78,715/QALY in Medicare adults, and $135,093/QALY in Medicaid adults
					  - Cost-effectiveness probability 0.997 overall
					    - Ranged from 0.667 for Medicaid adults to 0.98 for dual-eligible adults to 0.997 for Medicare adults

Hayes et al., 2019[35]	Cohort simulation modeling study, CBA	US, +1 year from analysis	575,408 Medicare FFS beneficiaries using data from 183 Medicare FFS beneficiaries
			- Age: 25% <65 yr, 14% 65–75 yr, 61% ≥75 yr
			- Sex: 63% female, 37% male
			- Race & ethnicity: 72% non-Hispanic White, 8% non-Hispanic Black, 12% Hispanic, and 9% other
			- Education: 35% less than high school, 34% high school, 30% more than high school
			- Income: 66% reporting income <$25,000, 34% reporting income ≥$25,000
			- Health: 100.0% with 1+ ADL deficits, 100.0% food insecurity, 100.0% with 2+ of the following conditions: CHF, depression, diabetes, emphysema, asthma, COPD, Alzheimer's, dementia, osteoporosis, other heart
				7 days of home-delivered MTMs following an inpatient stay (at a cost of $175.98/person/wk)	Readmissions
					- 9,719 fewer readmissions attributable to MTMs
					
					Costs
					- Total costs of $101,258,974
					- Gross savings due to decreased readmissions of $158,606,687
					- Net savings of $57,347,713
					  - Ranged from -$36,324,648 for individuals with one inpatient stay (-$100/person) to $61,152,345 for individuals with four or more inpatient stays ($951.18/person)
					- Savings to cost of 1:57, with every dollar spent on the program resulting in $1.57 in savings
					  - Additional savings likely from decreased ED and SNF admissions

			condition, paralysis, Parkinson's, rheumatoid arthritis, and stroke - Location: 67% metro area, 33% non-metro area - Marital status: 34% married, 33% widowed, 22% divorced or separated, 10% never married		
Observational Studies					

Raso, 2020[36]	Retrospective observational study (thesis), CEA	Detroit, MI, 2019	171 patients of the CHASS center, an FQHC
			- Mean age: 49 yr
			- Sex: primarily female
			- Race & ethnicity: 44% Black, 37% Hispanic or Latino, and 19% White, Native American, or other
			- Income: household income <$24,999
			- Health: 100.0% with DM
				16–18 weeks of free fruit and vegetable prescriptions, cooking demonstrations, healthy recipes, and standardized nutrition education	Health outcomes
					- Decrease in HbA1c of 0.532 (95% CI 0.313–0.751)
					  - Ranged from 0.576 (95% CI 0.342–0.811) in the 140 individuals who completed 2+ visits to 0.635 (95% CI 0.372–0.898) in the 122 individuals who completed 3+ visits to 0.768 (95% CI 0.481–1.054) in the 96 individuals who completed 4+ visits to 0.927 (95% CI 0.614–1.240) in the 27 individuals who completed 5+ visits
					
					Costs
					- Total cost of $59,152.66/year (95% CI $37,413.01–$88,451.38)
					  - Included $3,645.56 for screening and referral (95% CI $3,030.80–$4,322.29), $555.74 for transportation (95% CI $222.42–$1,081.74), $2,073.76 for capital and material farmers market costs (95% CI $2,425.37–$1,749.80), $9,255.85 for nutrition and cooking education costs (95% CI $4,012.02–$15,539.69), and $43,621.75 for general program costs (95% CI $22,467.27–$72,582.62)
					
					Cost effectiveness
					- ACER of $1,901/unit change in HbA1c (95% CI $1,208.72–$3,016.96)
					  - Labor costs (74%) significantly affected the ACER, with the director increasing weekly hours from 12 to 20 hours/week increasing the ACER from $1,901 to $3,295.11

Thomas et al., 2013[37]	Observational ecological study with modeled projections, CSA and BIA	US, 2009	392,594 additional adults eligible for MOW using data from 71,984 facility-year observations from 15,034 free-standing certified nursing homes operating for at least one year, augmented with data from State Program Reports 2005–2009, the Online Survey Certification and Reporting databases, the Minimum Data Set, the Area Resource Files of the	Increasing the number of adults who receive MOW by one percent	- Decrease of 0.2% in states' low-care nursing home population
					- Decrease of 1,722 low-care, dually eligible elderly adults in nursing homes
					  - Ranged from 3 in WY to 137 in NY
					- Initial costs of $117,568,07 in Title III-C2 expenditures
					  - Ranged from $139,563 in WY to $17,935,899 in CA
					- Initial Medicaid savings of $109 million
					  - Ranged from $173,196 in WY to $11,427,143 in NY
					- Annual state and federal savings in 26/50 states and annual costs in 22/50 states
					  - Highest savings $5,728,849 in PA, highest costs $11,472,019 in FL
					  - Savings increased to 46/50 states with FL and CA excluded

			HRSDA, and Medicare 2009 enrollment records		
Berkowitz et al., 2019[6]	Retrospective observational cohort study, CAA	MA, 2016–2019	1,020 adults (499 recipients, 521 nonrecipients) - Mean age: 52.7 ± 14.5 yr - Sex: 55.7% female - Race & ethnicity: 23.8% non-Hispanic White, 13.5% non-Hispanic Black, 4.5% Hispanic, 1.7% multiracial or other, 56.5% unknown - Poverty: 9.7% living in a zip code affected by poverty - Insurance: 56.3% Medicaid, 20.9% Medicare, 11.2% private payer, 11.7% other - Health: 17.6% with a disability, 16.2% HIV-positive, 37.5% history of cancer, 28.0% history of ESRD, 27.3% history of DM, 28.7% history of CHF, mean comorbidity index 4.23 ± 4.1 - Health care use: mean number of inpatient, SNF, and home health visits in the past year 1.0 ± 1.9, 0.5 ± 3.7, and 15.4 ± 64.3, respectively, mean health care costs in the past year $54,470 ± $73,081	Delivery of 10 MTMs/wk (cost of $350/person/wk including dietary tailoring, food, and delivery)	Inpatient and SNF admissions - MTMs associated with significantly fewer inpatient admissions (IRR of 0.51, 95% CI 0.22–0.80) and fewer estimated admissions (risk difference -519/1,000 person-yrs; 95% CI -360– -678/1,000 person-yrs) - MTMs associated with fewer SNF admissions (IRR of 0.28, 95% CI 0.0– -0.60) and fewer estimated admissions (risk difference -913/1,000 person-yrs; 95% CI -689– -1457/1,000 person-yrs) - Findings remained robust in sensitivity analyses Costs - MTMs associated with 16% lower health care costs, estimated at decrease of $712/mo (95% CI $1,930 decrease–$505 increase) - Had nonrecipients received MTMs, mean monthly costs would have decreased from $4,591 to $3,838 (risk difference of -$753, 95% CI -$280– -$1,255; relative risk difference of 0.84 mean per person per month expenditures, 95% CI 0.67–0.998) - Findings remained robust in sensitivity analyses

Berkowitz et al., 2018[38]	Retrospective cohort study, CAA	MA, 2014–2016	3,257 adults members of the CCA
			- 133 MTM recipients
			  - Mean age: 57.4 ± 8.4 yr
			  - Sex: 55.64% female
			  - Race & ethnicity: 37.59% non-Hispanic White, 20.30% non-Hispanic Black, 8.27% Hispanic, 33.84% Asian/other/multiracial/did not answer
			  - Percent of households living
				Delivery of 10 MTMs/wk + snacks (MTMs) for 6 months (at a cost of $350/person/wk)
				
				Delivery of 10 non-MTMs/wk (N-MTMs) for 6 months (at a cost of
					Health care utilization
					- MTMs associated with fewer ED visits (aIRR of 0.30, 95% CI 0.20–0.45), inpatient admissions (aIRR 0.48, 95% CI 0.26–0.90), and emergency transportation use (aIRR 0.28, 95% CI 0.16–0.51)
					- N-MTMs associated with fewer ED visits (aIRR 0.56, 95% CI 0.47–0.68) and emergency transportation use (aIRR 0.62, 95% CI 0.49–0.78) but not fewer inpatient admissions (aIRR 0.88, 95% CI 0.69–1.11)
					
					Costs
					- MTMs associated with lower health care costs, estimated at decrease of $570/mo (95% CI $208 decrease–$931 decrease)
					- MTM net savings of $220/mo

			in poverty: 16.26 ± 5.14 - Insurance: 81.95% One Care, 18.05% Senior Care Options - Health: comorbidity index 0.26 ± 0.39, 18.80% prescribed insulin, 59.40% prescribed an anti-hypertensive, 29.32% prescribed another CVD medication, 48.12% prescribed a PPI, 30.08% prescribed an inhaler, 25.56% prescribed oral steroids, 45.11% prescribed antibiotics - Health care use: average annual baseline health care costs $11,251 ± $8,553 - 1,002 MTM matched controls - Mean age: 57.9 ± 5.4 yr - Sex: 53.49% female - Race & ethnicity: 35.22% non-Hispanic White, 13.78% non-Hispanic Black, 8.27% Hispanic, 42.73% Asian/other/multiracial/did not answer - Percent of households living in poverty: 16.06 ± 3.24 - Insurance: 78.04% One Care, 21.96% Senior Care Options - Health: comorbidity index 0.26 ± 0.25, 10.24% prescribed insulin, 52.62% prescribed an anti-hypertensive, 26.14% prescribed another CVD medication, 30.39% prescribed a PPI, 15.18% prescribed an inhaler,	$146/person/ wk)	- N-MTMs associated with lower health care costs, estimated at a decrease of $156/mo (95% CI $5 decrease–$308 decrease) - N-MTM net savings of $10/mo - Findings remained robust in sensitivity analyses

			15.09% prescribed oral steroids, 33.22% prescribed antibiotics - Health care use: average annual baseline health care costs \$5,475 ± \$3,849 - 624 N-MTM recipients - Mean age: 73.5 ± 7.5 yr - Sex: 60.74% female - Race & ethnicity: 12.50% non-Hispanic White, 0.80% non-Hispanic Black, 28.53% Hispanic, 58.17% Asian/other/multiracial/did not answer - Percent of households living in poverty: 17.86 ± 6.66 - Insurance: 20.03% One Care, 79.97% Senior Care Options - Health: comorbidity index 0.17 ±0.32, 15.71% prescribed insulin, 70.35% prescribed an anti-hypertensive, 40.22% prescribed another CVD medication, 43.27% prescribed a PPI, 19.55% prescribed an inhaler, 14.90% prescribed oral steroids, 30.61% prescribed antibiotics - Health care use: average annual baseline health care costs \$5,446 ± \$5,619 - 1,318 N-MTM matched controls - Mean age: 73.1 ± 5.9 yr - Sex: 63.78% female - Race & ethnicity: 12.50% non-Hispanic White, 0.80% non-Hispanic Black, 28.53% Hispanic, 58.17%		

			Asian/other/multiracial/did not answer - Percent of households living in poverty: 17.96 ± 5.38 - Insurance: 22.55% One Care, 77.45% Senior Care Options - Health: comorbidity index 0.17 ± 0.26, 13.71% prescribed insulin, 68.11% prescribed an anti-hypertensive, 40.22% prescribed another CVD medication, 40.01% prescribed a PPI, 17.69% prescribed an inhaler, 16.37% prescribed oral steroids, 36.75% prescribed antibiotics - Health care use: average annual baseline health care costs $5,095 ± $3,887		

Gurvey et al., 2013[39]	Retrospective cohort study, CSA	Philadelphia, PA, 2008–2010	698 adult clients of the MANNA
			- 65 recipients
			  - Mean age: 52 ± 6.2 yr
			  - Sex: 42% female, 58% male
			  - Race & ethnicity: 77% Black, 20% White, 3% other; 98% non-Hispanic, 2% Hispanic
			  - Health: 43% with AIDS/HIV, 38% with chronic pulmonary disease, 26% with cancer, 25% with mild liver disease, 25% with diabetes without complications, 20% with CHF, 20% with renal disease
			- 633 nonrecipients
			  - Mean age: 51 ± 1.2 yr
			  - Sex: 36% female, 64% male
			  - Race & ethnicity: 79%
				21 MTMs/wk + MNT, nutrition counseling, and meal planning from RDs	Health care utilization
					- Significantly lower average monthly inpatient visits among recipients (0.2) compared to nonrecipients (0.4)
					- Significantly lower average monthly inpatient LOS among recipients (10.7) compared to nonrecipients (17.1)
					- Significantly higher average monthly ED visits among recipients (0.6) compared to nonrecipients (0.3)
					- Significantly higher percent of individuals discharging to home among recipients (93%) compared to nonrecipients (72%)
					
					Costs
					- Significantly lower average monthly health care costs among recipients ($28,268) compared to nonrecipients ($40,906)
					- Significantly lower average monthly inpatient costs among recipients ($132,441) compared to nonrecipients ($219,639)
					- Higher average monthly ED visit costs among recipients ($4,893) compared to nonrecipients ($3,700)

			Black, 19% White, 2% other; 99% non-Hispanic, 1% Hispanic - Health: 47% with AIDS/HIV, 35% with chronic pulmonary disease, 31% with cancer, 23% with mild liver disease, 22% with diabetes without complications, 19% with CHF, 17% with renal disease		
Martin et al., 2018	Quasi-experimental study, CSA	ME, 2013–2015	622 adult MMC patients - Mean age: 71.7 ± 13.1 yr - Insurance: 100.0% Medicare - Health: 100.0% designated by CMS as at high risk of readmission, 37.0% obese, 30.7% overweight, 22.0% normal weight, 3.5% underweight; 10.8% principal diagnosis of HF, 6.6% atrial fibrillation/dysrhythmias, 5.0% septicemia, 4.8% acute MI, 4.2% coronary atherosclerosis, 14.8% other; 92.2% discharged to home or home health care, 7.8% discharged to SNF, rehab, or other health setting	4 weeks of tools and skills training* by a nurse or social worker with up to 7 frozen MTMs (MTMs) (meal cost of $70 per person per 7 days) 4 weeks of tools and skills training* by a nurse or social worker without MTMs (N-MTMs) *training covered medication reconciliation and self-management, use of a patient-centered health record, follow-up with a PCP and specialist, and	Readmissions - 30-day readmission rate of 10.3% among MTM recipients (95% CI 8.1%–12.9%) compared to 12.3% among N-MTM recipients (95% CI 10.0%–15.2%) Costs - Cost savings of $212,160 - Return on investment of 387% - Benefit-cost ratio of $3.87 for every $1.00 spent

				understanding of clinical indicators of worsening condition and next steps	
Shan et al., 2017[41]	Retrospective cohort study, CAA	CA, GA, NC, OK, RI, TX; 2009–2014	14,019 Medicare FFS beneficiaries	MOW	Health care utilization - Hospitalization significantly decreased by 38.9%, 38.0%, and 31.0% at 30, 90, and 180 days post MOW enrollment - ED visits significantly decreased by 28.2%, 20.7%, and 12.7% at 30, 90, and 180 days post MOW enrollment - Nursing home admissions significantly decreased by 27.8%, 37.4%, and 25.3% at 30, 90, and 180 days post MOW enrollment Health care costs - Average per person hospitalization costs significantly decreased by $361.50, $1,154.80, and $1,355.60 at 30, 90, and 180 days post MOW enrollment - Average per person SNF costs significantly decreased by $244.30, $651.80, and $363.20 at 30, 90, and 180 days post MOW enrollment - Average per person ED costs significantly decreased by $22.36 and $42.57 at 30 and 90 days post MOW enrollment and decreased by $27.47 at 180 days post MOW enrollment Additional analysis comparing beneficiaries to matched controls showed controls had significantly higher rates of hospitalization, ED visits, and SNF admissions

Haddad et al., 2025[42]	Single-arm prospective cohort feasibility study, CSA	OH, 2021–2022	60 Cleveland Clinic patients
			- Median age: 63.5 yr (range 51–81)
			- Sex: 63.3% female, 36.7% male
			- Race & ethnicity: 70% Black, 25.0% White, 3.3% American Indian, 1.7% Asian, 1.7% Middle Eastern
			- Income: 75.0% annual income <$29,999, 20.0% between $30,000 and $49,999, 3.3% between $50,000 and $99,999, and 1.7% between $100,000 and $349,999; 100.0% living in a food desert
			- Insurance: 41.7% insured by
				14 home-delivered frozen MTMs/wk + 7 servings/wk of mild or milk substitute + fruit + bread needed to achieve 2/3 of daily recommended calorie intake for 3 months	Health care utilization
					- Significant decrease in ED visits 60 days before (0.650) vs. 60 days into (0.233), 90 days before (1.000) vs. 90 days into (0.200), 90 days into (0.200) vs. after (0.850), and 180 days before (1.700) vs. 180 days after (1.150) MTMs
					  - Increase in ED visits 30 days before (0.283) vs. 30 days into (0.383) and 60 days into (0.233) vs. 60 days after (0.517) MTMs
					  - Decrease or no change in ED visits 30 days before (0.283) vs. 30 days after (0.283, 60 days before (0.650) vs. 60 days after (0.517), 90 days before (1.000) vs. 90 days after (0.850), and 30 days into (0.383) vs. 30 days after (0.283) MTMs
					- Significant decrease in inpatient LOS 180 days before (5.067) vs. 180 days after (3.200) MTMs
					  - Increase in inpatient LOS 30 days before (0.233) vs. 30 days into (0.783), 30 days before (0.233) vs. 30 days after (0.317), 60 days before (0.683) vs. 60 days after (0.950), 60 days into (0.483) vs. 60 days after (0.950), and 90 days into (0.567) vs. 90 days after (1.300) MTMs
					  - Decrease in inpatient LOS 60 days before (0.683) vs. 60 days into (0.483), 90

			Medicare, 35.0% dual-eligible, 21.7% by Medicaid, 1.7% uninsured
			- Health: 100.0% diagnosed with CHF, uncontrolled DM, or uncontrolled HTN; 100.0% food insecure
			- Health care use: 100.0% with at least 2 ED or hospital visits in the prior 6 months
					days before (1.317) vs. 90 days into (0.567), 90 days before (1.317) vs. 90 days after (1.300), and 30 days during (0.567) vs. 30 days after (1.300) MTMs
					- No significant difference in standardized GPH or GMH scores between baseline and 30, 60, or 90 days after MTMs as assessed by PROMIS-10
					
					Costs
					- Average per patient health care savings of $12,046 between 180 days before vs. 180 days after MTMs
					- Cost savings achieved in 37/60 patients, with average cost savings of $29,172/patient
					- Total annual program costs of $189,000
					  - 42.9% of costs attributed to building, implementing, and maintaining the meal-ordering website, and 57.1% of costs attributed to making, packaging, and delivering the meals
					
					Feasibility
					  - Client and service outcomes ranked favorably, and upfront costs, penetration, and sustainability ranked poorly using the IOM Implementation Outcomes Framework

Hager et al., 2025[43]	Retrospective cohort study, CCA	Massachusetts, 2020–2023	22,511 MassHealth members eligible (i.e., with at least 1 health need and food insecurity) for nutrition-related benefits via the Flexible Services Program
			- 15,557 adult and 4,846 child recipients
			  - Mean age: 44.7 ± 12.6 yr (adults), 9.4 ± 4.9 yr (children)
			  - Sex: 63.5% female, 33.2% male, 3.3% missing (adults); 54.2% male, 41.7% female, 4.1% missing (children)
			  - Race & ethnicity: 33.8% Hispanic, 30.2% non-Hispanic White, 15.6% non-Hispanic Black, 15.6% missing, <5% Native American, <5% Asian/Pacific Islander (adults); 53.2% Hispanic, 18.7% missing, 16.0% non-Hispanic Black, 11.3%
				- Nutrition education (16 ACOs)
				- MTMs (14 ACOs)
				- Connection to other community and federal nutrition resources (14 ACOs)
				- PRx (10 ACOs)
				- Kitchen supplies (7 ACOs)
				- Food boxes or groceries (6 ACOs)
				- Home-delivered meals (4 ACOs)
					Health care utilization
					- Adults: Nutrition-related services associated with significant decreases in inpatient hospitalizations (aIRR 0.78, 95% CI 0.67–0.91) and ED visits (aIRR 0.85, 95% CI 0.78–0.94), but not primary care visits (aIRR 1.00, 95% CI 0.99–1.02)
					  - Adults with enrollments > 90 days: Nutrition-related services associated with significant decreases in inpatient hospitalizations (aIRR 0.75, 95% CI 0.64–0.87) and ED visits (aIRR 0.82, 95% CI 0.76–0.90), but not primary care visits (aIRR 1.00, 95% CI 0.98–1.01)
					- Children: Nutrition-related services associated with decreases in inpatient hospitalizations (aIRR 0.68, 95% CI 0.23–1.99), ED visits (aIRR 1.09, 95% CI 0.84–1.41), and primary care visits (aIRR 1.06, 95% CI 1.00–1.13)
					  - Children with enrollments > 90 days: Nutrition-related services associated with decreases in inpatient hospitalizations (aIRR 0.96, 95% CI 0.40–2.27) and ED visits (aIRR 1.11, 95% CI 0.85–1.45), and primary care visits (aIRR 1.04, 95% CI 0.99–1.09)
					
					Health care costs
					- Adults: Nutrition-related services associated with decreases in health care costs (-$1,178, 95% CI -$2,759–$402)
					  - Adults with enrollments > 90 days: Nutrition-related services associated with decreases in health care costs (-$2,502, 95% CI -$4,641–$363)
					- Children: Nutrition-related services associated with increases in health care costs ($234, 95% CI -$1,111–$1,579)
					  - Children with enrollments > 90 days: Nutrition-related services associated with

			non-Hispanic White, <5% Native American, <5% Asian/Pacific Islander (children) - Education: 34.4% high school, 23.3% missing, 17.2% less than high school, 12.7% some college/associates degree, 6.3% other, 4.8% bachelors degree or higher (adults); N/A (children) - Employment: 44.8% unemployed, 12.1% full time, 12.0% part time, 17.4% not in labor force, 11.7% prefer not to say or other, 2.1% student (adults); N/A (children) - Homelessness: 19.9% enrolled in the Flexible Services Housing Program, 17.0% at risk of homelessness, 7.8% homeless (adults); 9.8% enrolled in the Flexible Services Housing Program, 9.6% at risk of homelessness, 2.2% homeless (children) - Insurance: 100.0% ACO enrollees - Health: 62.9% with anxiety disorder, 45.5% with HTN, 45.4% with obesity, 45.4% with other mental health conditions, 34.1% with DM, 32.4% with tobacco use, 31.1% with disability, 30.8% with depression, 24.8% with asthma, 24.1% with drug abuse, 18.5% with bipolar disorder (adults); 34.4% with obesity, 27.4% with other		increases in health care costs ($729, 95% CI -$977–$2,436)

			mental health conditions, 26.2% with asthma (children) - Health care use (in the 6 months prior to enrollment): 0.7 ± 2.1 acute unplanned hospitalizations, 0.9 ± 1.9 ED visits, $15,503 ± $23,775 health care costs, 5.9 ± 3.4 primary care visits (adults); 0.1 ± 0.9 acute unplanned hospitalizations, 0.4 ± 0.8 ED visits, $6,337 ± $16,885 health care costs, 2.7 ± 2.9 primary care visits (children) - 1,497 adult and 611 child nonrecipients (eligible but did not enroll) - Median age: 43.2 ± 13.2 yr (adults), 9.5 ± 4.9 yr (children) - Sex: 59.4% female, 38.8% male, 1.8% missing (adults); 55.6% male, 44.0% female, 0.4% missing (children) - Race & ethnicity: 44.76% non-Hispanic White, 21.1% Hispanic, 16.4% non-Hispanic Black, 16.0% missing, <5% Native American, <5% Asian/Pacific Islander (adults); 30.1% non-Hispanic White, 29.6% missing, 22.6% non-Hispanic Black, 16.7% Hispanic, <5% Native American, <5% Asian/Pacific Islander (children) - Education: 43.1% missing, 27.4% high school, 8.9% less than high school, 8.8% some college/associates degree, 5.1% other, 3.7% bachelors		

			degree or higher (adults); N/A (children) - <u>Employment</u>: 33.3% unemployed, 26.6% prefer not to say or other, 10.7% full time, 10.7% part time, 14.4% not in labor force, 4.2% student (adults); N/A (children) - <u>Homelessness</u>: 12.1% enrolled in the Flexible Services Housing Program, 13.0% at risk of homelessness, 8.8% homeless (adults); 8.3% at risk of homelessness, 5.7% enrolled in the Flexible Services Housing Program, 2.4% homeless (children) - <u>Insurance</u>: 100.0% ACO enrollees - <u>Health</u>: 61.5% with anxiety disorder, 46.2% with other mental health conditions, 40.5% with obesity, 40.4% with tobacco use, 40.3% with HTN, 31.6% with disability, 31.4% with depression, 30.4% with drug abuse, 27.9% with DM, 24.3% with asthma, 23.6% with bipolar disorder (adults); 31.3% with obesity, 26.3% with other mental health conditions, 23.4% with asthma (children) - <u>Health care use (in the 6 months prior to recipient program enrollment)</u>: 0.7 ± 2.1 acute unplanned hospitalizations, 0.9 ± 1.9 ED visits, \$14,651 ± \$24,462 health care costs, 4.3 ± 4.6 primary care visits (adults);		

			0.1 ± 1.1 acute unplanned hospitalizations, 0.4 ± 0.8 ED visits, $5,740 ± $13,155 health care costs, 2.7 ± 3.1 primary care visits (children)		

ACER = average cost-effectiveness ratio, ACO = Accountable Care Organization, ADLs = activities of daily living, AIDS/HIV = acquired immunodeficiency syndrome/human immunodeficiency virus, aIRR = adjusted incidence rate ratio, AK = Alaska, AL = Alabama, BRFSS = Behavioral Risk Factor Surveillance System, CA = California, CCA = Commonwealth Care Alliance, CHASS = Community Health and Social Services, CHD = coronary heart disease, CHF = congestive heart failure, CI = confidence interval, CMS = Centers for Medicare and Medicaid Services, CO = Colorado, COPD = chronic obstructive pulmonary disease, CVD = cardiovascular disease, DM = diabetes mellitus, DOC-M = diabetes obesity cardiovascular disease microsimulation, ED = Emergency Department, EBT = electronic benefit transfer, ESRD = end-stage renal disease, FFS = fee-for-service, FL = Florida, FPL = federal poverty level, FQHC = Federally Qualified Health Center, GA = Georgia, GMH = global mental health, GPH = global physical health, HbA1c = hemoglobin A1c, HF = heart failure, HI = Hawaii, HTN = hypertension, IADLs = instrumental activities of daily living, ICER = incremental cost-effectiveness ratio, ID = Idaho, IOM = Institute of Medicine, IRR = incidence rate ratio, LOS = length of stay, MA = Massachusetts, MANNA = Metropolitan Area Neighborhood Nutrition Alliance, MD = Maryland, ME = Maine, MEPS = Medical Expenditure Panel Survey, MI (location) = Michigan, MI (health) = myocardial infarction, MMC = Maine Medical Center, MNT = medical nutrition therapy, mo = month, MOW = Meals on Wheels, MTM = medically tailored meal, NC = North Carolina, NH = New Hampshire, NHANES = National Health and Nutrition Examination Survey, NY = New York, OH = Ohio, OK = Oklahoma, PA = Pennsylvania, PCP = primary care provider, PPI = proton pump inhibitor, PROMIS-10 = patient-reported outcomes measurement information system, PRx = produce prescription, QALY = quality adjusted life year, RD = Registered Dietitian, RI = Rhode Island, SNAP = Supplemental Nutrition Assistance Program, SNF = skilled nursing facility, TX = Texas, UI = uncertainty interval, US = United States, wk = week, WTP = willingness-to-pay, WV = West Virginia, WY = Wyoming, yr = year